EXERCÍCIOS DE CÁLCULO DIFERENCIAL E INTEGRAL 1

Ficha 1

1

a) $\dfrac{\frac{x}{2}}{\frac{2}{x}} = \dfrac{x^2}{4} = \left(\dfrac{x}{2}\right)^2$, $x \neq 0$ //

b) $\dfrac{x+1}{\frac{1}{x}+\frac{1}{1}} = \dfrac{x+1}{\frac{x+1}{x}} = \dfrac{x(x+1)}{x+1} = x$ // $x \neq 0$

c) $\dfrac{1}{1+x} + \dfrac{1}{x^2+x} = \dfrac{x \times 1}{x(1+x)} + \dfrac{1}{x^2+x} = \dfrac{x+1}{x(x+1)} = \dfrac{1}{x}$, $x \neq 0$ //

d) $\sqrt{x^2} = \pm x$, $x \neq 0$ //

e) $(\sqrt{x})^2 = x$, $x \geq 0$ //

f) $4^x \cdot \dfrac{4}{2^x} = \dfrac{4^{(x+1)}}{2^x} = \dfrac{2^{2(x+1)}}{2^x} = 2^{2x+2-x} = 2^{x+2}$ //

g) $2^{x^2} \cdot (2^x)^2 = 2^{x^2} \cdot 2^{2x} = 2^{2x+x^2}$ //

h) $\dfrac{\sqrt[3]{x^2}}{\sqrt[6]{x}} = \dfrac{(x^2)^{1/3}}{(x)^{1/6}} = \dfrac{x^{2/3}}{x^{1/6}} = x^{\frac{2}{3}-\frac{1}{6}} = x^{\frac{3}{6}} = x^{\frac{1}{2}} = \sqrt{x}$ //

i) $\sqrt{x-2} \cdot \sqrt{x+2} = \sqrt{(x-2)\cdot(x+2)} = \sqrt{x^2+2x-2x-4} =$

$= \sqrt{x^2-4}$ // $\quad x^2-4 \geq 0 \Leftrightarrow x \geq \pm 2$

j) $\dfrac{\sqrt{x}}{\sqrt{x+1}-\sqrt{x}} = \dfrac{\sqrt{x}\cdot(\sqrt{x+1}+\sqrt{x})}{(\sqrt{x+1}-\sqrt{x})(\sqrt{x+1}+\sqrt{x})} = \dfrac{\sqrt{x(x+1)}+x}{x+1-x} =$

$= \sqrt{x^2+x}$ //

$x \geq 0$

$x+1 \geq 0 \Leftrightarrow x \geq -1$

$\sqrt{x+1} \neq \sqrt{x}$

k) $\log\left(\frac{1}{x}\right) + \log(x^2) = \log\left(x^{-1} \cdot x^2\right) = \log x$ //

l) $\log(2x^2 + 2x^{-2}) + \log\left(\frac{x^2}{2} + \frac{x^{-2}}{2}\right) =$

$= \log(2(x^2+x^{-2})) + \log\left(\frac{1}{2}(x^2+x^{-2})\right) =$

$= \log 2 + \log(x^2+x^{-2}) + \log 2^{-1} + \log(x^2+x^{-2}) =$

$= \log 2 - \log 2 + 2\log(x^2+x^{-2}) = 2\log(x^2+x^{-2})$

2

a) $(x^2-3x+2)(x-1) \geq 0$

$x-1=0 \Leftrightarrow x=1$ //

$x^2-3x+2=0 \Leftrightarrow x = \frac{3 \pm \sqrt{9-4.1.2}}{2.1} \Leftrightarrow x = \frac{3 \pm \sqrt{9-8}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{3 \pm \sqrt{1}}{2} \Leftrightarrow x = \frac{3+1}{2} \vee x = \frac{3-1}{2} \Leftrightarrow x = 2 \vee x = 1$ //

$-\infty$		1		2	$+\infty$	
$-$		0	$+$	$+$	$+$	$x-1$
$+$		0	$-$	0	$+$	x^2-3x+2
$-$		0	$-$	0	$+$	

$\Omega = [2; +\infty[\cup \{1\}$

b) $x \leq 2-x^2 \Leftrightarrow 0 \leq -x^2 - x + 2 \Leftrightarrow -x^2-x+2 \geq 0$ //

$-x^2-x+2=0 \Leftrightarrow x = \frac{1 \pm \sqrt{1-4.(-1).2}}{-2} \Leftrightarrow$

$\Leftrightarrow x = \frac{1 \pm \sqrt{9}}{-2} \Leftrightarrow x = \frac{1 \pm 3}{-2} \Leftrightarrow x = \frac{1+3}{-2} \vee x = \frac{1-3}{-2} \Leftrightarrow$

$\Leftrightarrow x = -\frac{4}{2} \vee x = \frac{2}{2} \Leftrightarrow x = -2 \vee x = 1$ //

	$-\infty$		-2		1	$+\infty$
	$-$	$\|$	0	$+$	0	$+$

$\Omega = [-2, 1]$

c) $x^2 \leq 2 - x^4 \Leftrightarrow 0 \leq -x^4 - x^2 + 2 \quad x^2 = y$

$-y^2 - y + 2 \qquad y = \frac{1 \pm \sqrt{1 - 4.(-1).2}}{-2} \Leftrightarrow y = \frac{1 \pm 3}{-2} \Leftrightarrow$

$x^2 \neq -2 \vee x^2 = 1 \qquad \Leftrightarrow y = \frac{-2}{-2} \vee y = \frac{4}{-2} \Leftrightarrow y = 1 \vee y = -2$

d) $x^3 + x \leq 2x^2 \Leftrightarrow \quad x^3 - 2x^2 + x \leq 0$

$\Leftrightarrow x(x^2 - 2x + 1) \leq 0$

$x = 0$ //

$x^2 - 2x + 1 = 0 \Leftrightarrow x = \frac{2 \pm \sqrt{4 - 4.1.1}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm \sqrt{4-4}}{2} \Leftrightarrow x = 1$ //

x	$-\infty$		0		1	$+\infty$
x	$-$	$\|$	0	$+$	$+$	$+$
$x^2 - 2x - 2$	$+$	$\|$	$+$	$+$	0	$+$
	$-$	$\|$	0	$+$	0	$+$

$\Omega =]-\infty, 0] \cup 1$

e) $\sqrt[3]{x^2 + 2x} = 2 \Leftrightarrow x^2 + 2x = 2^3 \Leftrightarrow x^2 + 2x - 8 = 0 \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm \sqrt{4 - 4.1.(-8)}}{2.1} \Leftrightarrow x = \frac{-2 \pm \sqrt{4 + 32}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm \sqrt{36}}{2} \Leftrightarrow x = \frac{-2 + 6}{2} \vee x = \frac{-2 - 6}{2} \Leftrightarrow x = 2 \vee x = -4$ //

f) $\sqrt[3]{x-1} = \sqrt{x-1} \Leftrightarrow (x-1)^{1/3} = (x-1)^{2/2} \Leftrightarrow \frac{1}{3} = \frac{1}{2}$ // P.F.

Esta equação não tem soluções $S = \emptyset$

g) $\frac{x-1}{x^2-1} \leq 1 \Leftrightarrow \frac{x-1}{(x+1)(x-1)} - 1 \leq 0 \Leftrightarrow \frac{1}{x+1} - \frac{x+1}{x+1} \leq 0 \Leftrightarrow$

$\Leftrightarrow \frac{1-(x+1)}{x+1} \leq 0 \Leftrightarrow \frac{x}{x+1} \leq 0$ //

$S =]-1, 0]$

	$-\infty$		-1		0		$+\infty$
x	$-$	$-$	$-$	$-$	0	$+$	$+$
$x+1$	$-$	$-$	0	$+$	$+$	$+$	$+$
	$+$	$+$	ND	$-$	0	$+$	$+$

h) $x = \frac{1}{x} \Leftrightarrow {}^2x = 1 \Leftrightarrow x = 1 \vee x = -1$ //

i) $x < \frac{1}{x} \Leftrightarrow x - \frac{1}{x} < 0 \Leftrightarrow \frac{x^2-1}{x} < 0$ //

$x^2 - 1 = 0 \Leftrightarrow x^2 = 1 \Leftrightarrow x = \pm 1$ //

	$-\infty$		-1		0		1	$+\infty$
x	$-$	$-$	$-$	$-$	0	$+$	$+$	$+$
x^2-1	$+$	$+$	0	$-$	$-$	$-$	0	$+$
	$-$	$-$	0	$+$	ND	$-$	0	$+$

$S =]-\infty, 0[\cup]0, 1[$

j) $x < |x| \Leftrightarrow$

$\Leftrightarrow x < x \vee x < -x \Leftrightarrow 2x < 0 \Leftrightarrow x < 0$

P.F.

$S =]-\infty, 0[$

k) $|x| \geq \frac{x}{2} + 1 \Leftrightarrow |x| \geq \frac{x}{2} + \frac{2}{2} \Leftrightarrow |x| \geq \frac{x+2}{2} \Leftrightarrow$

$\Leftrightarrow x \geq \frac{x+2}{2} \vee -x \geq \frac{x+2}{2} \Leftrightarrow 2x \geq x+2 \vee -2x \geq x+2 \Leftrightarrow$

$\Leftrightarrow 2x - x \geq 2 \vee -2x - x \geq 2 \Leftrightarrow x \geq 2 \vee -3x \geq 2 \Leftrightarrow$

$\Leftrightarrow x \geq 2 \vee x \leq -\frac{2}{3}$ // $S =]-\infty, -\frac{2}{3}] \cup [2, +\infty[$

l) $|x| \leq |x-2| \Leftrightarrow x \leq |x-2| \vee -x \leq |x-2| \Leftrightarrow$

$\Leftrightarrow x \leq x-2 \vee x \leq -x+2 \vee -x \leq x-2 \vee -x \leq -x+2 \Leftrightarrow$

$\Leftrightarrow 0 \leq -2 \vee 2x \leq 2 \vee -2x \leq -2 \vee 0 \leq 2 \Leftrightarrow$

P.F P.V

$\Leftrightarrow x \leq 1$ // $S =]-\infty, 1]$

m) $|x^2-2| \leq 2 \Leftrightarrow x^2-2 \leq 2 \vee -(x^2-2) \leq 2 \Leftrightarrow$

$\Leftrightarrow x^2 \leq 2+2 \vee -x^2 \leq 2-2 \Leftrightarrow$

$\Leftrightarrow x^2 \leq 4 \vee -x^2 \leq 0 \Leftrightarrow x \leq \pm\sqrt{4} \Leftrightarrow x \leq 2 \vee x \leq -2$ //

P.V.

$S =]-\infty, 2] \cup]-\infty, -2]$

n) $\frac{x^4-76}{|x-1|} \leq 0 \Leftrightarrow \frac{x^4-76}{x-1} \leq 0 \vee -\frac{x^4-76}{-x+2} \leq 0 \Leftrightarrow$

$x = 2 \vee x = -2$

$\Leftrightarrow \frac{(x^2-4)(x^2+4)}{x-1} \vee \frac{(x^2-4)(x^2+4)}{1-x}$

$x = 1$ $\quad x = -1$

$S =]-\infty, -2] \cup]1, 2]$

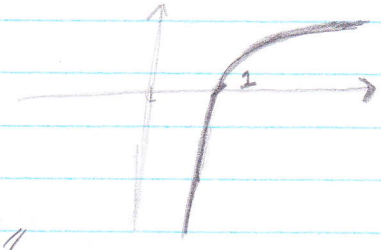
o) $e^{x^3} < 1 \Leftrightarrow e^{x^3} < e^0 \Leftrightarrow x^3 < 0$ // $S = \mathbb{R}^-$

p) $e^{-2x} - 2e^{-x} \leq -1 \Leftrightarrow e^{-2x} - 2e^{-x} + 1 \leq 0$

$e^{-x} = y \Rightarrow y^2 - 2y + 1 = 0 \Leftrightarrow y = \frac{2 \pm \sqrt{4-4.1.1}}{2} \Leftrightarrow$

$\Leftrightarrow y = \frac{2}{2} \Leftrightarrow y = 1$

$e^{-x} = 1 \Leftrightarrow e^{-x} = e^0 \Leftrightarrow -x = 0$ //



q) $\log\left(\frac{1}{x}\right) \geq 0 \Leftrightarrow$

$\Leftrightarrow \log 1 - \log x \geq 0 \Leftrightarrow -\log x \geq 0$ //

$S =]0, 1]$

r) $\log(x^2-3) \geq 0 \Leftrightarrow x^2-3 \geq 1 \Leftrightarrow x^2 \geq 4 \Leftrightarrow$

$\Leftrightarrow x \geq \pm 2$

$-\infty$		-2		2		$+\infty$
$+$	$+$	0	$-$	0	$+$	$+$

$\mathcal{S} =]-\infty, -2] \cup [2, +\infty[$

3

a) $\{x : \frac{x-1}{x+1} \leq 1\} \Leftrightarrow x \in \mathbb{R} \setminus \{-1\}$

b) $\{x : \frac{x^4-1}{x^3} \leq x\} \overset{x \in}{\Leftrightarrow} \mathbb{R} \setminus \{0\}$

$x^4-1 \leq x \cdot x^3 \Leftrightarrow x^4-1 \leq x^4$ P.V

c) $\{x : |3x-4| \geq x^2\} \overset{x \in}{\Leftrightarrow} [-4, 1]$ //

$3x-4 \geq x^2 \vee -3x+4 \geq x^2 \Leftrightarrow -x^2-3x-4 \geq 0 \vee -x^2-3x+4 \geq 0$

$-x^2+3x-4=0 \Leftrightarrow$

$x = \frac{-3 \pm \sqrt{9-(4 \cdot (-1) \cdot (-4))}}{-2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-3 \pm \sqrt{-5}}{-2}$ impossível

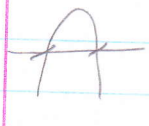
$-x^2-3x+4=0 \Leftrightarrow$

$\Leftrightarrow x = \frac{3 \pm \sqrt{9-(4 \cdot (-1) \cdot 4)}}{-2} \Leftrightarrow$

$\Leftrightarrow x = \frac{3 \pm \sqrt{9+16}}{-2} \Leftrightarrow x = \frac{3 \pm \sqrt{25}}{-2} \Leftrightarrow$

$\Leftrightarrow x = \frac{3+5}{-2} \vee x = \frac{3-5}{-2} \Leftrightarrow$

$\Leftrightarrow x = \frac{8}{-2} \vee x = \frac{-2}{-2} \Leftrightarrow x = -4 \vee x = 1$ //



d) $\{x : |x-1|(x^2-4) \geq 0\} \overset{x \in}{\Leftrightarrow}]-\infty, -2] \cup \{1\} \cup [2, +\infty[$

$(x-1)(x^2-4) \geq 0 \vee (x+1)(x^2-4) \geq 0$

$x-1=0 \vee x^2-4=0 \vee x+1=0 \vee x^2-4=0 \Leftrightarrow$

$\Leftrightarrow x = 1 \vee x = \pm\sqrt{4} \vee x = -1 \vee x = \pm\sqrt{4}$

	$-\infty$		-2		-1		1		2	$+\infty$
$\lvert x-1\rvert$	+	+	+	+	+	+	0	+	+	+
x^2-4	+	+	0	−	−	−	−	−	0	+
	+	+	0	−	−	−	0	−	0	+

$S =]-\infty, -2] \cup \{1\} \cup [2, +\infty[$

e) $\{x : (|x|-1)(x^2-4) \leq 0\} \Leftrightarrow x \in]-\infty, -2] \cup [-1, 2]$

C.A. $(|x|-1)(x^2-4) \leq 0 \Leftrightarrow (x-1)(x^2-4) \leq 0 \vee$

$\vee (-x-1)(x^2-4) \leq 0$

$(x-1)=0 \vee x^2-4=0 \vee -x-1=0 \vee x^2-4=0 \Leftrightarrow$

$\Leftrightarrow x=1 \vee x=\pm 2 \vee x=-1 \vee x=\pm 2$ //

$S =]-\infty, -2] \cup [-1, 2]$

f) $\{x : |x^2-1| \leq |x+1|\} \Leftrightarrow x \in [-1, 2]$

$x^2-1 \leq x+1 \vee x^2-x \leq -x-1 \vee -x^2+1 \leq x+1 \vee -x^2+1 \leq -x-1 \Leftrightarrow$

$\Leftrightarrow x^2-x-2 \leq 0 \vee x^2+x \leq 0 \vee -x^2+x \leq 0 \vee -x^2+x+2 \leq 0$

$x^2+x=0 \Leftrightarrow x^2=-x \Leftrightarrow x=-1$ //

$-x^2-x=0 \Leftrightarrow -x^2=x \Leftrightarrow x=-1$ //

$x^2-x-2=0 \Leftrightarrow x=\frac{1 \pm \sqrt{1-(4.1.(-2))}}{2} \Leftrightarrow \frac{1 \pm \sqrt{9}}{2}=x \Leftrightarrow x=\frac{1+3}{2} \vee$

$\vee x=\frac{1-3}{2} \Leftrightarrow$

$\Leftrightarrow x=2 \vee x=-1$ //

$-x^2+x+2=0 \Leftrightarrow x=\frac{-2\pm\sqrt{1-(4\cdot(-2)\cdot 2)}}{-2} \Leftrightarrow x=\frac{-1\pm\sqrt{9}}{-2} \Leftrightarrow$

$\Leftrightarrow x=\frac{-1+3}{-2} \vee x=\frac{-1-3}{-2} \Leftrightarrow x=-1 \vee x=2$ //

	$-\infty$		-1		2		$+\infty$
x^2+x	+	+	0	+	+	+	+
x^2-x-2	+	+	0	−	0	+	+
	+	+	0	−	0	+	+
$-x^2-x$	−	−	0	−	−	−	−
$-x^2+x+2$	−	−	0	+	0	−	−
	+	+	0	−	0	+	+

g) $\{x : x^2-|x|-2\leq 0\} \Leftrightarrow x\in[-2,2]$

$x^2-x-2\leq 0 \quad \vee \quad x^2+x-2\leq 0$

$x^2-x-2=0 \Leftrightarrow x=\frac{1\pm\sqrt{1-(4\cdot 1\cdot(-2))}}{2} \Leftrightarrow x=\frac{1\pm\sqrt{1+8}}{2} \Leftrightarrow$

$\Leftrightarrow x=\frac{1\pm 3}{2} \Leftrightarrow x=2 \vee x=-1$ // $\quad x>0$

$x^2+x-2=0 \Leftrightarrow x=\frac{-1\pm\sqrt{1-(4\cdot 1\cdot(-2))}}{2} \Leftrightarrow x=\frac{-1\pm 3}{2} \Leftrightarrow$

$\Leftrightarrow x=-2 \vee x=1$ // $\quad x<0$

	$-\infty$		-2		-1		1		2		$+\infty$
x^2-x-2	+	+	+	+	0	−	−	−	0	+	+
x^2+x-2	+	+	0	−	−	−	0	+	+	+	+

h) $\{x : \frac{x}{|x|-1} \geq 0\} \Leftrightarrow x \in \mathbb{R}_0^+ \setminus \{1\}$

$\frac{x}{|x|-1} \geq 0 \quad , \quad x \neq 1 \text{ e } x \neq -1$

i) $\{x : \frac{x^2 - |x|}{x-3} \leq 0\} \Leftrightarrow x \in]-\infty, -1] \cup \{0\} \cup [1, 3[$

$x - 3 = 0 \Leftrightarrow x = 3$

$x^2 - |x| = 0 \Leftrightarrow x^2 - x = 0 \wedge x \geq 0 \vee x^2 + x = 0 \wedge x < 0 \Leftrightarrow$

$\Leftrightarrow x^2 = x \wedge x \geq 0 \vee x^2 = -x \wedge x < 0 \Leftrightarrow x = 1 \vee x = -1$

	$-\infty$		-1		0	1		3		$+\infty$
$x^2 - \lvert x \rvert$	+	+	0	-	0	0	+	+	+	+
$x - 3$	-	-	-	-	-	-	-	0	+	+
	-	-	0	+	0	0	-	ND	+	+

Ficha 2

1.

a) $\{1\} \subset \{1, \{2,3\}\}$ falso

b) $\{1\} \in \{1, \{2,3\}\}$ falso

c) $2 \in \{1, \{2,3\}\}$ falso

d) $1 \in \{\mathbb{R}\}$ falso 1 não é elemento de $\{\mathbb{R}\}$

e) $\emptyset = \{x \in \mathbb{N} : x = x+1\}$ Verdadeira - $x = x+1$ nunca se verifica

f) $\emptyset \in \{0\}$ falso

g) $\emptyset \subset \{0\}$ Verdadeiro

h) $\forall_{x \in \mathbb{R}} \; x > 0 \Leftrightarrow x^{-1} > 0$ Verdadeira

i) $\forall_{x \in \mathbb{R}} \; x > 1 \Leftrightarrow x^{-1} < 1$ Verdadeira - se $x > 1$ então $\frac{1}{x} < 1$

j) $\forall_{x,y \in \mathbb{R}} \; x < y \Rightarrow y^{-1} < x^{-1}$ Verdadeira

K) $\forall_{x \neq 0} \; x^2 > 0$ Verdadeira - a função x^2 nunca é negativa

l) $\forall_{x,y \in \mathbb{R}} \; x < y \Rightarrow x^2 < y^2$ falso

m) $\forall_{x,y \in \mathbb{R}} \; x < y < 0 \Rightarrow x^2 > y^2$ Verdadeira

2

$\forall_{a>0}\ a + \frac{1}{a} \geq 1$

Se $a \geq 1$

se $a = 1$ $\quad 1 + 1 \geq 1$

$1 \leq \frac{1}{a} < 2$, $a > 0$

então

para qualquer $a > 0$

$a + \frac{1}{a} \geq$ é verdade

se $a < 1 \Rightarrow \frac{1}{a} > 1$

então $a + \frac{1}{a} \geq 1$ cqd

3

a) $1 + 3 + \ldots + (2n-2) = n^2$, $\forall n \in \mathbb{N}_1 = P(n)$

1) provar que é verdade para $n = 1$

$(2 \times 1 - 1) = 1 = 1^2$, como queríamos demonstrar

2) $P(n) \Rightarrow P(n+1)$

$$\overbrace{1 + 3 + \ldots + (2n-2)}^{n^2} + (2(n+1) - 1) = (n+1)^2 \Leftrightarrow$$

$$\Leftrightarrow n^2 + (2n + 2 - 1) = n^2 + 2n + 1 \Leftrightarrow$$

$$\Leftrightarrow n^2 + 2n + 1 = n^2 + 2n + 1$$, cqd

Apartir de 1) e 2) conclui-se que $P(n)$ é verdadeira para todo o $n \in \mathbb{N}_1$

b) $\frac{1}{1.2} + \frac{1}{2.3} + \ldots + \frac{1}{n(n+1)} = \frac{n}{n+1}$, $\forall n \in \mathbb{N}_1$

1) $n = 1$

$\frac{1}{1(1+1)} = \frac{1}{1+1} \Leftrightarrow \frac{1}{2} = \frac{1}{2} \Rightarrow$ cqd

2) Se $p(n) \Rightarrow p(n+1)$

$$\frac{1}{1\cdot 2} + \frac{1}{2\,3} + \ldots + \frac{1}{n(n+2)} + \frac{1}{(n+1)(n+1+1)} = \frac{n+2}{n+1+1} \Leftrightarrow$$

$$\Leftrightarrow \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} = \frac{n+2}{n+2} \Leftrightarrow$$

$$\Leftrightarrow \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{(n+1)(n+2)}{(n+2)(n+2)} \Leftrightarrow$$

$$\Leftrightarrow \frac{n^2+2n+1}{(n+2)(n+2)} = \frac{n^2+2n+2}{(n+1)(n+2} \quad // \text{ cqd}$$

apartir de 1) e 2) conclui-se que a proposição é verdadeira para todo o $n \in \mathbb{N}_1$

c) $(n!)^2 > 2^n n^2$, $\forall n \geq 4 \in \mathbb{N}$

1) $\boxed{n=4}$

$(4!)^2 > 2^4 \cdot 4^2 \Leftrightarrow 48^2 > 76.76 \Leftrightarrow$

$\Leftrightarrow 48^2 > 76^2$ cqd

2) $p(n) \Rightarrow p(n+1)$

$((n+2)!)^2 > 2^{n+1} \cdot (n+1)^2$

Pela hipótese: $(n!)^2 > 2^n n^2 \Leftrightarrow (n!)^2 (n+1)^2 > 2^n n^2 (n+1)^2 \Leftrightarrow$

$\Leftrightarrow (n!)^2 (n+1)^2 > 2^{n+1} n^2 (n+1)^2 \cdot \frac{1}{2} \Leftrightarrow$

$\Leftrightarrow (n!)^2 (n+1)^2 > 2^{n+1} (n+1)^2 \cdot \underline{\frac{n^2}{2}}$ $\quad \frac{n^2}{2} \geq 1 \Leftrightarrow$

$\Leftrightarrow n^2 \geq 2 \Leftrightarrow n \geq \sqrt{2}$, $n \geq 4$ cqd

d) $n! \geq 2^{n-1}$, $\forall n \in \mathbb{N}_1$

1) $\boxed{n=1}$

$1! \geq 2^{1-1} \Leftrightarrow 1 \geq 2^0 \Leftrightarrow 1 \geq 1$ cqd

2) se $p(n) \Rightarrow p(n+1)$

$(n+1)! \geq 2^{(n+1)-1} = 2^n$

Pela hipotese

$(n!)(n+1) \geq 2^{n-1}(n+2) \Leftrightarrow (n!)(n+1) \geq 2^n \cdot 2 \cdot \frac{(n+2)}{2}$

$\Leftrightarrow (n!)(n+1) \geq 2^n \quad \frac{n+1}{2} \rightarrow \geq 1$ (V) $\quad \frac{n+1}{2} \geq 1 \Leftrightarrow n+1 \geq 2$

$\Leftrightarrow n \geq \frac{2}{1} \Leftrightarrow n \geq 1$ cqd

4

a) $1+2+3+\ldots+n = \frac{n(n+1)}{2}$ $\forall n \in \mathbb{N}_1$

$\boxed{n=1}$

$1 = \frac{1(1+1)}{2} \Leftrightarrow 1 = 1$ cqd

$\boxed{P(n) \Rightarrow P(n+1)}$

$1+2+3+\ldots+n+(n+1) = \frac{(n+1)(n+2+1)}{2} \Leftrightarrow$

$\Leftrightarrow \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2} \Leftrightarrow$

$\Leftrightarrow \frac{n(n+2) + 2(n+1)}{2} = \frac{(n+1)(n+2)}{2} \Leftrightarrow$

$\Leftrightarrow \frac{n^2+n+2n+2}{2} = \frac{n^2+2n+n+2}{2}$ cqd //

b) $(a-1)(1+a+\ldots+a^n) = a^{n+1}-1 \quad \forall a \in \mathbb{R} \quad \forall n \in \mathbb{N}$

n=1

$(a-1)(a+1) = a^{1+1}-1 \iff a^2 + a - a - 1 = a^2 - 1$ // cqd

Se P(n) ⇒ P(n+1)

$(a-1)(1+a+\ldots+a^n+a^{n+1}) = a^{n+2}-1 \iff$

h

$\iff a^{n+1} - 1 + (a-1)\cdot a^{n+1} = a^{n+2} - 1 \iff$

$\iff a^{n+1} - 1 + a^{n+2} - a^{n+1} = a^{n+2} - 1$ cqd

hipotese de indução:

c) $\frac{1}{2!} + \frac{2}{3!} + \ldots + \frac{n}{(n+2)!} = 1 - \frac{1}{(n+1)!} \quad \forall n \in \mathbb{N}$

n=1 $\sum_{i=1}^{n} \frac{i}{(i+1)!}$

$\frac{1}{2!} = 1 - \frac{1}{(1+1)!} \iff \frac{1}{2} = 1 - \frac{1}{2} \iff \frac{1}{2} = \frac{2}{2} - \frac{1}{2} \iff$

$\iff \frac{1}{2} = \frac{1}{2}$ cqd

Se P(n) ⇒ P(n+2)

$\underbrace{\frac{1}{2!} + \frac{2}{3!} + \ldots + \frac{n}{(n+1)!}} + \frac{n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!} \iff$

h

$\iff 1 - \frac{1}{(n+1)!} + \frac{n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!} \iff$

$\iff 1 - \frac{1}{(n+1)!} + \frac{n+1}{(n+1)!(n+2)} = 1 - \frac{1}{(n+2)!} \iff$

$\Leftrightarrow 1 - \frac{-(n+2)+n+2}{(n+1)!\,(n+2)} = 1 - \frac{1}{(n+2)!} \Leftrightarrow$

$\Leftrightarrow 1 + \frac{-n-2+n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!} \Leftrightarrow$

$\Leftrightarrow 1 - \frac{1}{(n+2)!} = 1 - \frac{1}{(n+2)!}$ cqd

d) $1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 \quad n \in \mathbb{N}_1$

n = 1

$1^3 = \left(\frac{1(1+1)}{2}\right)^2 \Leftrightarrow 1 = 1^2$ cqd

Se p(n) ⇒ p(n+1)

$\underbrace{1^3 + 2^3 + 3^3 + \dots + n^3} + (n+1)^3 = \left(\frac{(n+1)(n+2)}{2}\right) \Leftrightarrow$

$\overset{h}{\Leftrightarrow} \left(\frac{n(n+1)}{2}\right)^2 + (n+1)^3 = \left(\frac{(n+1)(n+2)}{2}\right)^2 \Leftrightarrow$

$\Leftrightarrow \frac{n^2}{4} + n + 1 = \frac{(n+2)^2}{4} \Leftrightarrow \frac{n^2+4n+4}{4} = \frac{n^2+4n+4}{4}$ cqd

e) $1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n} \quad \forall n \in \mathbb{N}_1$

n = 1

$1 \geq \sqrt{1} \Leftrightarrow 1 \geq 1$ cqd

Se $p(n) \Rightarrow p(n+1)$

$$\frac{1}{\sqrt{n+1}}\left(\sqrt{n}+\frac{1}{\sqrt{n+1}}\right)=\frac{\sqrt{n}}{\sqrt{n+1}}+\frac{1}{n+2}=$$

$$\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\dots+\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{n+1}} \geq \sqrt{n}+\frac{1}{\sqrt{n+1}} \qquad \left(\geq \sqrt{n}\right)$$

$$=\frac{\sqrt{n}\sqrt{n+1}+1}{n+2} \geq 1 \quad \times\sqrt{n+1} \geq \sqrt{n+1}$$

$$\frac{\sqrt{n}+1}{\sqrt{n+1}}=\frac{\sqrt{n}+\sqrt{n+1}+1}{\sqrt{n+1}} \geq \frac{\sqrt{n}\sqrt{n+1}}{\sqrt{n+1}}=\frac{n+1}{\sqrt{n+1}}=\sqrt{n+1}$$

$$1+\sqrt{n}\sqrt{n+1} \geq \sqrt{n}\sqrt{n+1}+1$$

$a \geq b,\ c>0$
$a \cdot c \geq b \cdot c$

5

a) $(n+2)! \geq 2^{2n} \qquad \forall n \in \mathbb{N}_1$

$n=1$

$(1+2)! \geq 2^{2 \cdot 1} \Leftrightarrow 3! \geq 2^2 \Leftrightarrow 6 \geq 4$ // cqd

Se $p(n) \Leftrightarrow p(n+1)$

$(n+1+2)! \geq 2^{2(n+1)} \Leftrightarrow (n+3)! \geq 2^{2n+2} \Leftrightarrow$

$\Leftrightarrow (n+2)! \cdot (n+3) \geq 2^{2n+2}$ //

pela hipótese: $(n+2)! \geq 2^{2n} \Leftrightarrow (n+2)!\,(n+3) \geq 2^{2n}(n+3) \Leftrightarrow$

$\Leftrightarrow (n+2)!\,(n+3) \geq 2^{2n} \cdot 2^2 \cdot \frac{(n+3)}{2^2} \Leftrightarrow (n+2)!\,(n+3) \geq 2^{n+2} \cdot \frac{n+3}{4}$

$\frac{n+3}{4} \geq 1 \Leftrightarrow n+3 \geq 4 \Leftrightarrow n \geq 4-3 \Leftrightarrow n \geq 1$ cqd

b) $2n-3 < 2^{n-2} \qquad n \in \mathbb{N} \wedge n \geq 5$

$n=5$

$2 \cdot 5-3 < 2^{5-2} \Leftrightarrow 7 < 2^3 \Leftrightarrow 7 < 8$ // cqd

Se $p(n) \Leftrightarrow p(n+1)$

$2(n+1) - 3 < 2^{n-1} \Leftrightarrow 2n - 1 < 2^{n-1}$

$2n - 3 < 2^{n-2} \Leftrightarrow 2n - 3 + 2 < 2^{n-1} + 2 \Leftrightarrow 2n - 1 < \left(2^{n-1} + 2\right)\frac{1}{2}$

$\Leftrightarrow 2n - 1 < 2^{n-2} + 1$ cqd

c) $7^n - 1$ é multiplo de 6 $\forall n \in \mathbb{N}_1$

↳ $6K$

$n = 1$

$7^1 - 1 = 6 = 6.1 \Rightarrow$ cqd

Se $p(n) \Leftrightarrow p(n+1)$

$7^{n+1} - 1 = 7 \cdot 7^n - 1$

d) $2^{2n} + 2$ é multiplo de 3 $\forall n \in \mathbb{N}_1$

$n = 1$

$2^{2\times 1} + 2 = 2^2 + 2 \Leftrightarrow 4 + 2 = 6$ cqd

↳ multiplo de 3

Se $p(n) \Leftrightarrow p(n+2)$

6 $a > -1$, $n \in \mathbb{N}$

$$(1+a)^n \geq 1 + n.a$$

$\boxed{n=1}$

$1+a \geq 1+a$ cqd

$\boxed{\text{Se } p(n) \Rightarrow p(n+1)}$

$$(1+a)^{n+1} \geq 1 + an + a$$

$(1+a)^n \geq 1 + n.a \Leftrightarrow (1+a)^n (1+a) \geq (1+na)(1+a)$

$\Leftrightarrow (1+a)^{n+1} \geq 1 + an + a + \underline{na^2}$ → ?

7. $P(n)$: "n^2+3n+1 é par"

a) $p(n+1): (n+1)^2 + 3(n+1) + 1 = n^2 + 1 + 3n + 3 + 1$

$= n^2 + 3n + 1 + 4$ é

pela hipótese "n^2+3n+1 é par"

e 4 é par, logo $n^2 + 3n + 1 + 4$ é par

b) $\boxed{n=1}$ $\quad 1^2+3.1+1=5$

não, $\exists n$ que torna falsa a condição $p(n)$

c) n^2+3n+1 é impar $\forall n \in \mathbb{N}$

$\boxed{n=1}$

$1^2+3.1+1=5$ 5 é impar cqd

$\boxed{P(n) \Rightarrow P(n+1)}$

$(n+1)^2+3(n+1)+1 = n^2+2n+1+3n+3+1 =$
$= n^2+3n+1+2n+4$

Pela hipotese n^2+3n+1 é impar

$2n+4$ é par, logo $n^2+3n+1+2n+4$ é impar

8. $f: \mathbb{N} \longrightarrow \mathbb{N}$

$f(0)=1$
$f(n+1)=(2n+2)(2n+1)f(n)$

Mostre que $f(n)=(2n)! \Rightarrow f(n+1)=(2(n+1))!=(2n+2)!$

$f(n+1)=(2n+2)(2n+1)(2n)!=(2n+2)!$ cqd

9.

$$(u_n) = \begin{cases} u_1 = 3 \\ u_{n+1} = \dfrac{3u_n}{(n+1)^2} \end{cases}$$

Mostre que $u_n = \dfrac{3^n}{(n!)^2} \quad \forall n \in \mathbb{N}_1$

$$u_{(n+1)} = \frac{3 \cdot \frac{3^n}{(n!)^2}}{(n+1)^2} = \frac{3 \cdot 3^n}{((n+1)!)^2} = \frac{3^{n+1}}{((n+1)!)^2} \quad \text{cqd}$$

10.

$$u_n \begin{cases} u_1 = 1 \\ u_{n+1} = \sqrt{2u_n^2 + 1} \end{cases}$$

Mostre que $u_n = \sqrt{2^n - 1}$

$\Downarrow$

$u_{n+1} = \sqrt{2^{n+1} - 1}$

$$u_{n+1} = \sqrt{2(\sqrt{2^n - 1})^2 + 1} = \sqrt{2(2^n - 1) + 1} =$$

$$= \sqrt{2 \cdot 2^n - 2 + 1} = \sqrt{2^{n+1} - 1} \quad \text{cqd}$$

11

n impar $\rightarrow 2n+1$

$(2n+1)^2 = 2n^2 + 4n + 1$

par

$n^2 \rightarrow 2n$

12.

Ficha 3

1.

$$A = \{ x \in \mathbb{R} : |x| \geq \frac{x}{2} + 2 \}$$

$$B = [-3, 4] \qquad C = \mathbb{R} \setminus \mathbb{Q}$$

a)

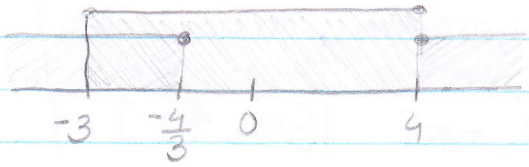
$$|x| \geq \frac{x}{2} + 2 \Leftrightarrow \underbrace{x \geq \frac{x}{2} + 2}_{x \geq 0} \vee \underbrace{x \leq -\frac{x}{2} - 2}_{x < 0} \Leftrightarrow$$

$$\Leftrightarrow x - \frac{x}{2} \geq 2 \vee x + \frac{x}{2} \leq -2 \Leftrightarrow \frac{2x - x}{2} \geq 2 \vee \frac{2x + x}{2} \leq -2 \Leftrightarrow$$

$$\Leftrightarrow \frac{x}{2} \geq 2 \vee \frac{3x}{2} \leq -2 \Leftrightarrow x \geq 4 \vee 3x \leq -4 \Leftrightarrow x \geq 4 \vee x \leq -\frac{4}{3}$$

$$A = \{ x : x \in]-\infty, -\frac{4}{3}] \cup [4, +\infty[\}$$

$$A \cap B = [-3, -\frac{4}{3}] \cup \{4\}$$



b)

Sup A → não existe
min(A∩B) → -3
max(A∩B) → 4
inf(A∩B∩C) → -3
sup(A∩B∩C) → -4/3
min(A∩B∩C) → Não existe

$]-3, -\frac{4}{3}[\cap (\mathbb{R} \setminus \mathbb{Q})$

2.

$$A = \{ x \in \mathbb{R} : \frac{x - 1}{x \log x} > 0 \} = x \in \mathbb{R}^+ \setminus \{1\}$$

$$x \log x = 0 \Leftrightarrow x = 0 \vee \log x = 0 \Leftrightarrow x = 0 \vee x = 1$$

$$B = \{ x \in \mathbb{R} : x = -\frac{1}{n}, n \in \mathbb{N}_1 \} = -1 \geq x > 0$$

	A	A∪B	
sup	NE	NE	$\mathbb{R}^+$ não tem supremo
inf	1	-1	
max	NE	NE	$\mathbb{R}^+$ não tem máximo
min	NE	-1	

3.

$[1/2, 2] \setminus 1$

$A = \{ x \in \mathbb{R} : \frac{1}{\log x} \geq 1 \}$ $\quad B = \{ 1 - \frac{(-1)^n}{n} , n \in \mathbb{N}_1 \}$

$\frac{1}{\log x} \geq 1 \Leftrightarrow$

$\frac{1}{\log x} - 1 \geq 0 \Leftrightarrow \frac{1 - \log x}{\log x} \geq 0$ | $1 - \log x = 0 \Leftrightarrow 1 = \log x \Leftrightarrow x = e$ | $\log x = 0 \Leftrightarrow x = 1$

	0		1		e		$+\infty$
$1 - \log x$	ND	+	+	+	0	-	
$\log x$	ND	-	0	+	+	+	
		-	ND	+	0	-	

$]1, e] = A$

	majorantes	minorantes	sup	inf	max	min
A	$[e, +\infty[$	$]-\infty, 1]$	e	1	e	—
B	$[2, +\infty[$	$]-\infty, 1/2]$	2	1/2	2	1/2

4.

$A = \{ x \in \mathbb{R} : x^2 + 2|x| > 3 \} = x \in]-\infty, -1[\cup]1, +\infty[$

a)

$x \geq 0 \qquad x < 0$

$x^2 + 2|x| > 3 \Leftrightarrow x^2 + 2 \cdot x > 3 \vee x^2 + 2 \cdot (-x) > 3$

$\Leftrightarrow x^2 + 2x - 3 > 0 \vee x^2 - 2x - 3 > 0$ //

$x > 1 \vee x < -3 \qquad x < -1 \vee x > 3$

$x^2 + 2x - 3 = 0 \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm \sqrt{4 - (4 \cdot 1 \cdot (-3))}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm \sqrt{4 + 12}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 \pm 4}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2 + 4}{2} \vee x = \frac{-2 - 4}{2} \Leftrightarrow$

$\Leftrightarrow x = 1 \vee x = -3$ //

$x^2 - 2x - 3 = 0 \Leftrightarrow$

$\Leftrightarrow x = \frac{2 \pm \sqrt{4 - (4 \cdot 1 \cdot (-3))}}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{2 \pm 4}{2} \Leftrightarrow x = \frac{2 + 4}{2} \vee x = \frac{2 \cdot 4}{2} \Leftrightarrow$

$\Leftrightarrow x = 3 \vee x = -1$ //

	$-\infty$		-3		-1		0		1		3		$+\infty$
$x^2 + 2x - 3$	+	+	0	-	-	-	-	-	0	+	+	+	
$x^2 - 2x - 3$	+	+	+	+	0	-	-	-	-	-	0	+	

b) $\inf A = NE \to]1, \sqrt{2}[$ $\quad B =]0, \sqrt{2}[$

$\min A \cap B = NE$

$\max A \cap B = NE$

$\max A \cap B \cap \mathbb{Q} = NE$

$\inf A \cap B \cap \mathbb{Q} = 1$ $\quad C = \{\sqrt{2} - \frac{1}{n} : n \in \mathbb{N}_1\}$

$\max C = NE$ $\quad 0 < C < \sqrt{2}$

$\max B \setminus C = NE$ $\quad \to \sup = \sqrt{2}$

5

a) $A = \{x \in \mathbb{R} : |x-1| \leq x^2 - 1\} =]-\infty, -1]$ $\quad B = [-2, 2]$

$|x-1| \leq x^2 - 1 \Leftrightarrow x - 1 \leq x^2 - 1 \Leftrightarrow x - 1 \geq 1 - x^2 \Leftrightarrow$

$\Leftrightarrow x \leq x^2 \vee x^2 + x \geq 1 + 1 \Leftrightarrow -x^2 + x \leq 0 \vee x^2 + x - 2 \geq 0$

$-x^2 + x = 0 \Leftrightarrow x^2 = x \Leftrightarrow x = \pm\sqrt{x} \Leftrightarrow x = \pm 1$ //

	$-\infty$	-1		1		2	$+\infty$
$-x^2+x$	$-$	0	$+$	0	$-$	$-$	$-$
x^2-x-2	$+$	0	$-$	$-$	$-$	0	$+$

Erro na passagem para a tabela: x^2+x-2 passa a x^2-x-2

$x^2 - x - 2 = 0 \Leftrightarrow x = \frac{1 \pm \sqrt{1 - (4 \cdot 1 \cdot (-2))}}{2} \Leftrightarrow x = \frac{1 \pm 3}{2} \Leftrightarrow x = \frac{4}{2} \vee x = -\frac{2}{2}$

$\Leftrightarrow x = 2 \vee x = -1$ //

b) $\max A \cap B = -2$ $\quad A \cap B =$

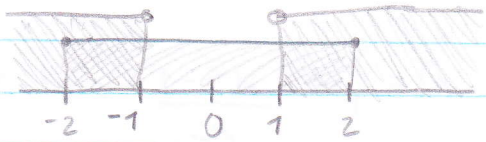
$\min A \cap B = 2$

$\sup (A \cap B) \setminus \mathbb{Q} = 2$

$\inf (A \cap B) \setminus \mathbb{Q} = -2$

$\max (A \cap B) \setminus \mathbb{Q} = 2$

$\min (A \cap B) \setminus \mathbb{Q} = -2$



6

a) $A=\{x: |x^2-2| \leq 2x+1\}$ $B=\mathbb{Q}$ $C=\left\{\frac{1}{k^2} : k\in\mathbb{N}\right\}$

$|x^2-2| \leq 2x+1 \Leftrightarrow x^2-2 \leq 2x+1 \vee x^2-2 \geq -2x-1 \Leftrightarrow$
$\Leftrightarrow x^2-2x-2-1\leq 0 \vee x^2+2x-2+1 \geq 0 \Leftrightarrow$
$\Leftrightarrow x^2-2x-3\leq 0 \vee x^2+2x-1\geq 0$ //

$x^2-2x-3=0 \Leftrightarrow x=\frac{2\pm\sqrt{4-(4\cdot 1\cdot(-3))}}{2} \Leftrightarrow x=\frac{2\pm\sqrt{4+12}}{2} \Leftrightarrow$
$\Leftrightarrow x=\frac{2\pm\sqrt{16}}{2} \Leftrightarrow$
$\Leftrightarrow x=\frac{2+4}{2} \vee x=\frac{2-4}{2} \Leftrightarrow x=3 \vee x=-2$ //

$x^2+2x-1=0 \Leftrightarrow x=\frac{-2\pm\sqrt{4-(4\cdot 1\cdot(-1))}}{2} = x=\frac{-2\pm\sqrt{8}}{2} \Leftrightarrow$
$\Leftrightarrow x=\frac{-2\pm 2\sqrt{2}}{2} \Leftrightarrow x=-1\pm\sqrt{2}$

	$-\infty$		$-1-\sqrt{2}$		-2		$-1+\sqrt{2}$		3	$+\infty$
x^2-2x-3	+	+	+	0	-	-	-	0	+	
x^2+2x-1	+	0	-	-	-	0	+	+	+	

b)

	$A\cap B$	C
sup	NE	1
inf	$-1\pm\sqrt{2}$	NE
max	NE	1
min	$-1\pm\sqrt{2}$	NE

7

a) $A=\left\{x: \frac{x^2-1}{x} \geq |x-1|\right\}$ $B=\{x: \sin x=0\}$ $C=\mathbb{N}$

$\frac{x^2-1}{x} \geq |x-1| \Leftrightarrow \frac{x^2-1}{x} \geq x-1 \vee -\left(\frac{x^2-1}{x}\right) \leq x-1 \Leftrightarrow$
$\Leftrightarrow x^2-1 \geq x(x-1) \vee -x^2+1 \leq x(x-1) \Leftrightarrow$
$\Leftrightarrow x^2-1 \geq x^2-x \vee -x^2+1 \leq x^2-x \Leftrightarrow$
$\Leftrightarrow x-1\geq 0 \vee -2x^2+x+1\leq 0$ //

$x-1=0 \Leftrightarrow x=1$ $-2x^2+x+1=0 \Leftrightarrow x=\frac{-1\sqrt{1-(4\cdot(-2)\cdot 1)}}{-4} \Leftrightarrow$

$\Leftrightarrow x=\frac{-1\pm 3}{-4} \Leftrightarrow x=\frac{2}{-4} \vee x=\frac{-4}{-4} \Leftrightarrow x=-\frac{1}{2} \vee x=1$ //

	$-\infty$		$-1/2$			1	$+\infty$
$\lvert x-1\rvert$	+		+		+	0	+
$-2x^2+x+1$	−		0		+	0	−
	−		0		+	0	−

b)

	majorantes	minorantes
$A \cap C$	$\mathbb{N}E$	$]-\infty, -1/2]$
$B \cap C$		

$\sup A = \mathbb{N}E$

$\inf A \cap C = -1/2$

$\min A \cap C = -1/2$

$\min B = \mathbb{N}E$

$\sup B \cap C = \mathbb{N}E$

8

a) $A = \{x : x \geq 0 \wedge \frac{x^4-4}{\lvert x-1\rvert} \leq 0\} \Leftrightarrow x = [0, \sqrt{2}] \setminus \{1\}$

$\frac{x^4-4}{\lvert x-1\rvert} \leq 0 \Leftrightarrow \frac{x^4-4}{x-1} \leq 0 \vee \frac{x^4-4}{1-x} \leq 0$

$x^4-4=0 \Leftrightarrow (x^2-2)^2=0 \Leftrightarrow x^2-2=0 \Leftrightarrow x = \pm\sqrt{2}$

	$-\infty$		$-\sqrt{2}$			0		1		$\sqrt{2}$	$+\infty$
x^4-4	+		0	−		−		−	−	0	+
$\lvert x-1\rvert$	+		+	+		+	+	0	+	+	+
	+		0	−		−	−	NE	−	0	+
x	−	−		−	−	0	+	+	+	+	+

b) $B = \{x : x \geq 0 \wedge \exists_{k \in \mathbb{N}} kx \notin \mathbb{Q}\}$

	A	$A \setminus B$
sup	$\sqrt{2}$	NE
inf	0	0
max	$\sqrt{2}$	NE
min	0	0

9

a) $A=\left\{x : \frac{x^2-2}{|x|-1} \leq 0\right\}$

$\frac{x^2-2}{|x|-1} \leq 0 \Leftrightarrow \frac{x^2-2}{x-1} \leq 0 \vee \frac{x^2-2}{-x-1} \leq 0$

$x^2-2=0 \Leftrightarrow x^2=2 \Leftrightarrow x=\pm\sqrt{2}$ //

$x-1=0 \Leftrightarrow x=1$ //

$-x-1=0 \Leftrightarrow -x=1 \Leftrightarrow x=-1$ //

	$-\infty$	$-\sqrt{2}$		-1		1		$\sqrt{2}$	$+\infty$
x^2-2	+	0	-	-	-	-	-	0	+
$\lvert x\rvert-1$	+	+	+	0	-	0	+	+	+
	+	0	-	NE	+	NE	-	0	+

$R=[-\sqrt{2}, -1[\cup]1, \sqrt{2}]$

b)

	$A \cap \mathbb{Q}$	$B \cap \mathbb{Q}$
sup	$\sqrt{2}$	NE
max	$\sqrt{2}$	NE
inf	$-\sqrt{2}$	$\sqrt{2}$
min	$-\sqrt{2}$	$\sqrt{2}$

$B=\left\{2^{n/2} : n \in \mathbb{N}_1\right\}$

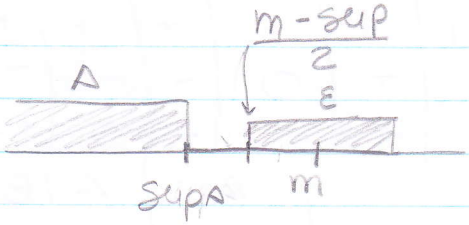
10. $A \subset \mathbb{R}$

A é majorado

$A \neq 0$

m é majorantes de A

$m \neq \sup A$



$V_\varepsilon(m)=\left\{x \in \mathbb{R} : |x-m|<\varepsilon\right\}$

Se $\varepsilon = \frac{m-\sup A}{2}$, temos que $\forall x \in A \; x \leq \sup(A)$

portanto $x < m-\varepsilon$ e $x \in V_\varepsilon(m)$ $\quad \sup(A) < m-\varepsilon \Leftrightarrow$

$\Leftrightarrow \sup(A) < m - \frac{(m-\sup A)}{2} \Leftrightarrow$

$\Leftrightarrow \frac{m-\sup A}{2} < m-\sup A$

11.

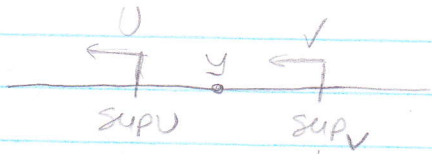
12. $U, V \neq \emptyset$
$U, V \in \mathbb{R}$ $\quad \sup U < \sup V$
U, V majorados

a) $\forall x \in U \Rightarrow x < \sup V$

$\forall x \in U \leqslant \sup U < \sup V \Rightarrow \forall x \in U < \sup V$

b) $\exists y \in V : y > \sup U$

$y \leqslant \sup V$ não é majorante que V $\Rightarrow \exists y \in V > \sup V$



$\sup V$ não há majorantes menores

Ficha 4

1ez	majorada	minorada	limitada	monótona
a)	— SIM	— SIM	— SIM	decrescente
b)	sim	sim (0)	sim	não
c)	não	não	não	não
d)	não	sim (0)	não	não
e)	não	sim (1)	não	sim crescente
f)	não	não	não	não
g)	— NÃO	— sim (0)	— NÃO	sim crescente

b) $\frac{n+(-1)^n}{n}$ $\{0, \frac{3}{2}, \frac{2}{3}, \dots\}$

min: $\frac{1+(-1)^1}{1} = 0$ sup = 2

a) $\frac{1}{\sqrt{n+1}}$

c) $(-1)^n n^2$ $\{-1, 2, -3, 4 \dots\}$

$\sqrt{n+1}$ → aumenta

então $\frac{1}{\sqrt{n+1}}$ diminui

min $= (-1)^1 1 = -1$

d) $n^{(-1)^n}$

min: $1^{(-1)^1} = -1$

$u_{2n} = 2n^{-1^{2n}} = 2n^{-1} = 2n$

$u_{2n+1} = (2n+1)^{-2} = 1/2n+2$

e) $1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n}$

$min = \frac{1}{2^0} = \frac{1}{1} = 1$

g) $u_1 = 0$

$u_{n+1} = \frac{2u_n+1}{3}$

$\{0, \frac{1}{3}, \frac{5}{6} \dots\}$

F) $u_1 = 1$, $u_{n+2} = -2u_n$ $\{1, -2, 4, -8 \dots$

$\lim_{x \to a} f(x) = b$ $\quad F(x) \longrightarrow B$

3

a) $\frac{1}{\sqrt{n+1}} \to 0$

$\forall \varepsilon > 0 \; \exists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p \Rightarrow |u_n - 0| < \varepsilon$

Dado $\varepsilon > 0$: $\left| \frac{1}{\sqrt{n+1}} - 0 \right| < \varepsilon \Leftrightarrow \frac{1}{\sqrt{n+1}} < \varepsilon \Leftrightarrow \frac{1}{\varepsilon} < \sqrt{n+1} \Leftrightarrow$

$\Leftrightarrow \left(\frac{1}{\varepsilon}\right)^2 < n+1 \Leftrightarrow n > \frac{1}{\varepsilon^2} - 1$

$\hookrightarrow = p$ cqd

b) $\frac{n^2}{n^2+1} \longrightarrow 1$

$\forall \varepsilon > 0 \; \exists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p \Rightarrow |u_n - 1| < \varepsilon$

dado um $\varepsilon > 0$: $\left| \frac{n^2}{n^2+1} - 1 \right| < \varepsilon \Leftrightarrow \frac{n^2 - n^2 + 1}{n^2+1} < \varepsilon \Leftrightarrow$

$\Leftrightarrow \frac{1}{n^2+1} < \varepsilon \Leftrightarrow \frac{1}{\varepsilon} < n^2 + 1 \Leftrightarrow \frac{1}{\varepsilon} - 1 < n^2 \Leftrightarrow$

$\Leftrightarrow \sqrt{\frac{1}{\varepsilon} - 1} < n$

$\hookrightarrow p$ cqd

c) $u_n = n^2$ é divergente $\longrightarrow$ não tende para um número real a

$\forall \varepsilon > 0 \; \exists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p \Rightarrow |n^2 - a| < \varepsilon$

dado um $\varepsilon > 0$:

$|n^2 - a| < \varepsilon \Leftrightarrow n^2 < \varepsilon - a \Leftrightarrow n < \sqrt{\varepsilon - a}$

$\hookrightarrow \nexists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p$

logo é divergente

4 a) $\frac{2n-1}{n+1} \to 2$

$\forall \varepsilon > 0 \; \exists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p \Rightarrow |u_n - 2| < \varepsilon$

Dado $\varepsilon > 0$:

$$\left|\frac{2n-1}{n+1} - \frac{2}{1}\right| < \varepsilon \Leftrightarrow \left|\frac{2n-1-2(n+1)}{n+1}\right| < \varepsilon \Leftrightarrow$$

$$\Leftrightarrow \left|\frac{2n-2-2n-2}{n+1}\right| < \varepsilon \Leftrightarrow \left|\frac{-3}{n+1}\right| < \varepsilon \Leftrightarrow \frac{3}{\varepsilon} < n+1 \Leftrightarrow$$

$$\Leftrightarrow \frac{3}{\varepsilon} - 1 < n \quad , \Rightarrow \frac{3}{\varepsilon} - 1 = p \quad \text{cqd}$$

b) $\frac{\sqrt{n^2-1}}{n} \to 1$

$\forall \varepsilon > 0 \; \exists p \in \mathbb{N} \; \forall n \in \mathbb{N} : n \geq p \Rightarrow |u_n - 1| < \varepsilon$

Dado um $\varepsilon > 0$:

$$\left|\frac{\sqrt{n^2-1}}{n} - 1\right| < \varepsilon \Leftrightarrow \left|\frac{\sqrt{n^2-1} - n}{n}\right| < \varepsilon \Leftrightarrow$$

$$\Leftrightarrow \sqrt{n^2-1} - n < \varepsilon . n \Leftrightarrow \sqrt{n^2-1} < \varepsilon n + n \Leftrightarrow$$

$$\Leftrightarrow n^2 - 1 < (\varepsilon n + n)^2 \Leftrightarrow n^2 - 1 < (\varepsilon n)^2 + 2n + n^2 \Leftrightarrow$$

$$\Leftrightarrow 1 < n(\varepsilon n + 2) \Leftrightarrow n > 1 \vee \varepsilon n + 2 > 1 \Leftrightarrow$$

$$\Leftrightarrow n > 1 \vee \varepsilon n > 1 - 2 \Leftrightarrow n > 1 \vee n > -\frac{1}{\varepsilon} \quad //$$

$\Rightarrow p = 1$ cqd

↳ $-\frac{1}{\varepsilon} < 0 \neq p$

5.

a) $\frac{(2n+1)^3+n}{n^3+1} = \left(\frac{+\infty}{+\infty}\right)$

$$\lim \frac{\left(\frac{2n+1}{n}\right)^3 + \frac{n}{n^3}}{\frac{n^3}{n^3} + \frac{1}{n^3}} = \lim \frac{\left(\frac{2n+1}{n}\right)^3 + \frac{1}{n^2}}{1+\frac{1}{n^3}} = \frac{2^3+0}{1+0} = 8$$

C.N.

$$\lim \frac{2n+1}{n} = \lim \frac{2n}{n} + \frac{1}{n} = \lim 2 + \frac{1}{n} = 2$$

b) $\lim \frac{(2n+1)^3+n^2}{(n+1)^2(n+2)} = \lim \frac{\left(\frac{2n+1}{n}\right)^3 + \frac{n^2}{n^3}}{\frac{n^3+4n^2+5n+2}{n^3}} =$

$$\lim \frac{\left(\frac{2n+1}{n}\right)^3 + \frac{1}{n}}{1+\frac{4}{n}+\frac{5}{n^2}+\frac{2}{n^3}} = \frac{2^3+0}{1+0} = 2^3 //$$

$(n^2+2n+1)(n+2) = n^3+2n^2+2n^2+4n+n+2 =$

$= n^3+4n^2+5n+2$

c) $\lim \frac{(n+1)^2+2n^4}{(n+1)^4+2n^2} = \lim \frac{\left(\frac{n+1}{n^2}\right)^2 + \frac{2n^4}{n^4}}{\left(\frac{n+1}{n}\right)^4 + \frac{2n^2}{n^4}} =$

$$= \lim \frac{\left(\frac{1}{n}+\frac{1}{n^2}\right)^2 + 2}{\left(1+\frac{1}{n}\right)^4 + \frac{2}{n^2}} = \frac{0+2}{1+0} = 2$$

d) $\lim \frac{\sqrt{n}-1}{\sqrt{n}-1} = \frac{1-1}{1} = 0$

$$\lim \frac{n-1}{n} = \frac{1-0}{1} = 1$$

$$\frac{n}{n+1} = \frac{1}{1+0} = 1$$

e) $\lim \frac{1}{n}\left(2+\frac{1}{n}\right) = 0$

F) $\lim \frac{1}{n}(2n+\sqrt{n}) \overset{0.\infty}{=} \lim \frac{2n}{n} + \frac{\sqrt{n}}{n} = \lim 2 + \frac{\sqrt{n}}{n} = 2$ ($\frac{\sqrt{n}}{n} \to 0$)

g) NE $S = \left\{\frac{-1}{n!}, \frac{1}{n!}\right\}$, logo é divergente

S: conjunto dos sublimites da sucessão

h) $\lim \frac{\sqrt{n}}{\sqrt[4]{4n^2+1}} = 1$ ($\sqrt[4]{4n^2+1} \to 1$)

$\frac{x_{n+1}}{x_n} = \frac{n+2}{n} \overset{\lim}{=} 1 + \frac{1}{n} = 1$

$\frac{\sqrt[4]{(n+2)^2+2}}{\sqrt[4]{n^2+2}} = \frac{\sqrt[4]{(n^2+2n+4)+2}}{\sqrt[4]{n^2+2}}$

i) $\lim \frac{\sqrt[3]{n+1}}{\sqrt[3]{n+2}} = \frac{\frac{\sqrt[3]{n^2}}{n^2} + \frac{8n}{n^2} + \frac{9}{n^2}}{\frac{\sqrt[3]{n^2}}{n^2} + \frac{2}{n^2}} = \frac{1 + 8/n + 9/n^2}{1 + 2/n^2}$

$= \frac{1+1}{1} = 2$

$\frac{n+2}{n+2} = \frac{1+1/n}{1+2/n} = 1$ $\qquad = \frac{1+0+0}{1+0} = 1$

j) $\lim \frac{n+1}{n!} = \lim \frac{n}{n!} + \frac{1}{n!} = 0$

k) NE $S = \left\{\frac{1}{\sqrt{n}}, 0\right\}$

S: Conjunto dos sublimites da sucessão

l) $\lim \frac{2^{n+1}+3^{n+1}}{2^n+3^n} = \lim \frac{2\left(\frac{2}{3}\right)^n + \frac{3^{n+1}}{3^n}}{\left(\frac{2}{3}\right)^n + 1} = \frac{2\cdot 0+3}{0+1} = 3$

($\left(\frac{2}{3}\right)^n \to 0$, $\frac{3^{n+1}}{3^n} \to 3$)

m) $\lim \frac{\sqrt[n]{1000}+1000}{n} = \frac{1+1000}{+\infty} = 0$

($\sqrt[n]{1000} \to 1$, $n \to +\infty$)

n) $\lim \frac{n^n}{n^n+1} = \lim \frac{n^n/n^n}{n^n/n^n + 1/n^n} = \frac{1}{1+0} = 1$

o) $\lim \frac{\sqrt[n]{3}}{\sqrt[3]{n}} = \frac{1}{1} = 1$

($\sqrt[n]{3} \to 1$, $\sqrt[3]{n} \to +\infty$)

p) $\lim \frac{4^n}{1+4^{n^2}} = \lim \frac{\frac{4^n}{4^{n^2}}}{\frac{1}{4^{n^2}} + \frac{4^{n^2}}{4^{n^2}}} = \lim \frac{4^{n-n^2}}{\left(\frac{1}{4}\right)^{n^2}+1} = 0$

($n-n^2 \to -\infty$, $\left(\frac{1}{4}\right)^{n^2} \to 0$)

q) $\lim \frac{(a^n)^2}{a^{n^2}} = \lim \frac{a^{2n}}{a^{n^2}} = \lim a^{2n-n^2} = 0$

6

$$x_n = \frac{a^n}{2^{1+2n}}$$

a) $\frac{a^n}{2^{1+2n}} = \frac{a^n}{2\cdot 2^{2n}} = \frac{1}{2}\left(\frac{a}{4}\right)^n$

$\lim a^n$ { $+\infty$, $a>1$; 1, $a=1$; 0, $-1<a<1$; NE, $a\le -1$ } ← Convergente

$-1 < \frac{a}{4} < 1 \Leftrightarrow -4 < a < 4$

b) $\left(\frac{a}{4}\right) = -1 \Leftrightarrow a = -4$

7

a) $u_n = \frac{1}{n} - 1$ tem termos em $]-\infty, 1[$ e é crescente

$u_n = -1$

b) $u_n = \frac{(-1)^n}{n}$ ~~monótona~~ convergente

$\frac{1 + (-2)^n}{n}$

c) $u_n = (-1)^n$ divergente, $|u_n|$ convergente

d) $u_n = (-1)^n$ limitada e divergente

e) $u_n = \frac{1}{n} \cdot (-2)^n$ $u_{2n} = 1/2$ e $u_{2n+2} = 1/3$ termos em $\frac{1}{n}$ e é divergente

f) $u_n = \frac{\sqrt{2}}{n}$ termos em $\mathbb{R} \setminus \mathbb{Q}$, $\to \mathbb{Q}$

8

$A = \{x \in \mathbb{R} : x^2 + 2|x| > 3\} =]-\infty, -1[\cup]1, +\infty[$

$B =]0, \sqrt{2}[$ $C = \{\sqrt{2} - \frac{1}{n} : n \in \mathbb{N}_1\}$

a)

b)

c)

d)

e)

f)

q)

$$\begin{cases} u_1 = a \\ u_{n+1} = r + u_n \end{cases} \qquad \begin{cases} v_1 = a \\ v_{n+1} = r\,v_n \end{cases}$$

a)

$u_n = a + (n-1)r$ | $v_n = ar^{n-1}$

$u_1 = a + (1-1)r = a$ | $v_1 = ar^{1-1} = a$

$u_{n+1} = a + (n+1-1)r = a + nr$ | $v_{n+1} = ar^n$

$u_{n+1} = r + u_n = r + a + (n-1)r =$ | $v_{n+1} = r\,v_n =$

$= r + a + nr - r =$ | $= r \cdot ar^{n-1} =$

$= a + nr$ cqd | $= \frac{r \cdot a \cdot r^n}{r} = ar^n$ cqd

b)

i) $\forall\, r, a > 0$

ii) $\forall a$, $\forall\, r < 0$

iii) $\forall a, r > 0$

iv) Se a ou r forem negativos (e apenas um deles)

c) u_n não é limitada $\forall a \in \mathbb{R}, r \neq 0$

$\begin{cases} u_1 = a \\ u_{n+1} = r + u_n \end{cases}$ $u_2 = r + a$ $u_n = (n-1)r + a$ se $n > 1 \Rightarrow n - 1 > 0$

Se $r > 0$ $(n-2) \cdot r > 0$ → $+\infty$

u_n não é limitada

Se $r < 0$ $(n-2)\,r < 0$ → $-\infty$

quando v_n é limitada? e convergente?

$v_n = a \cdot r^{n-1}$

limitada	convergente
Se a	se $a > 0$

10. $u_n \begin{cases} u_1 = 1 \\ u_{n+1} = 1 + \frac{u_n}{2} \end{cases}$

a) $u_n \leq 2 \quad \forall n \in \mathbb{N}_1$

$n=1$

$u_n = 1 \leq 2$

$1 + \frac{u_n}{2} \leq 2 \Leftrightarrow \frac{u_n}{2} \leq 1 \Leftrightarrow u_n \leq 1.2 \Leftrightarrow u_n \leq 2$ cqd

b) $u_1 = 1$

$u_2 = 1 + \frac{1}{2} = \frac{3}{2}$

$u_3 = 1 + \frac{3}{2} = \frac{5}{2}$

$u_4 = 1 + \frac{5}{2} = \frac{7}{2}$ ...

$u_1 < u_2 < u_3 < \dots < u_n$

logo é crescente

c) Se u_n é crescente e limitada por $1 \leq u_n \leq 2$ então $\lim u_n \longrightarrow 2$

11. $u_n \begin{cases} u_1 = \frac{3}{2} \\ u_{n+1} = \frac{u_n^2 + 2}{3} \end{cases}$ $\quad 1 < u_n < 2$

a) $u_1 = \frac{3}{2} \quad 1 < \frac{3}{2} < 2$

$P(n+2) \Rightarrow P(n)$: $1 < \frac{u_n^2 + 2}{3} < 2 \Leftrightarrow 3 < u_n^2 + 2 < 6 \Leftrightarrow$

$\Leftrightarrow 1 < u_n^2 < 4 \Leftrightarrow \sqrt{1} < u_n < \sqrt{4} \Leftrightarrow$

$\Leftrightarrow 1 < u_n < 2$ // cqd

b) u_n estritamente decrescente ≤ 0 (ver a))

$$u_{n+2} - u_n = \frac{u_n^2 + 2}{3} - u_n = \frac{u_n^2 + 2 - 3u_n}{3} \leq 0$$

p.p. $x^2 - 3x + 2 \leq 0 \Leftrightarrow x = \frac{3 \pm \sqrt{9 - 4.1.2}}{2} \Leftrightarrow x = \frac{3 \pm \sqrt{9-8}}{2} \Leftrightarrow$ $+ \; (\; - \;) \; +$

$\Leftrightarrow x = 3 \pm 1/2 \Leftrightarrow x = 2 \vee x = 1$ c.q.d.

c) u_n é decrescente e limitada, logo u_n é convergente. u_{n+2} é subsuc de u_n, então $\lim u_{n+1} = \lim u_n$

$a = \frac{a^2 + 2}{3} \Leftrightarrow a = 1 \vee a = 2$ Como u_n decrescente $\lim u_n = 1$

12. $u_1 > 1$ $\quad u_{n+1} = 2 - \frac{1}{u_n}$, $n \in \mathbb{N}_1$

$u_2 = 2 - \frac{1}{u_1} \Rightarrow u_2 > u_1$

$\hookrightarrow u_1 > 1$

logo, u_n é crescente

quando $n \to +\infty$, $u_n \to +\infty$

logo

$2 - \frac{1}{+\infty} = 2 = \lim u_n$

13.

a) $u_n : \begin{cases} u_1 = 1 \\ u_{n+1} = \sqrt{2 + u_n} \end{cases}$

PROVE:

$1 \leq u_n < 2 \quad \forall n \in \mathbb{N}$.

$\boxed{n=1}$ $u_1 = 1$ $\quad 1 \leq u_1 < 2$ c.q.d

n+1 $\quad 1 \leq \sqrt{2+u_n} < 2$

$1 \leq u_n < 2 \Leftrightarrow 2+1 \leq u_n+2 < 2+2 \Leftrightarrow$

$\Leftrightarrow \sqrt{3} \leq \sqrt{u_n+2} < \sqrt{4} \Leftrightarrow \sqrt{3} \leq \sqrt{u_n+2} < 2$

$1 \leq \sqrt{3} < 2$ cqd

b)

$u_1 = 1$

$u_2 = \sqrt{2+1} = \sqrt{3}$ $\quad u_1 < u_2 < u_3 \ldots < u_n$

$u_2 = \sqrt{2+\sqrt{3}}$

$\rightarrow u_n$ é crescente

$$u_{n+2} - u_n = \frac{(2-u_n)(u_n+1)}{u_n + \sqrt{2+u_n}}$$

$$u_{n+1} - u_n = \sqrt{2+u_n} - u_n = \frac{(\sqrt{2+u_n} - u_n)(\sqrt{2+u_n} + u_n)}{u_n + \sqrt{2+u_n}}$$

$$= \frac{2+u_n + u_n\sqrt{2+u_n} - u_n\sqrt{2+u_n} - u_n^2}{u_n + \sqrt{2+u_n}} = \frac{-u_n^2 + u_n + 2}{u_n}$$ cqd

c) u_n é crescente ✓ $\quad$ logo u_n é convergente

u_n é limitada ✓

14. $u_n > 0$ $\quad \frac{u_{n+1}}{u_n} < 1 \Rightarrow u_n$ é decrescente

decrescente $\quad \rightarrow u_{n+1} > u_n$

logo é limitada

$\rightarrow$ logo é convergente

15. $x_n \rightarrow$ monótona
$y_n \rightarrow$ limitada

$\rightarrow$ limitada?

mesmo limite?

$\forall n \in \mathbb{N}_1 \quad |u_n - y_n| < \frac{1}{n}$

limitada

Mesmo grau ⇒ limite é = ao produto das coeficientes de maior grau

$\frac{(-1)^n \cdot 1}{|a|^n} \to 0$ ↳ tende p/ 0

fichas

1. $u_n = \frac{(-1)^{n+1}}{n}$

- limitada
- $-\frac{1}{2} \leq u_n \leq 1$

$v_n = \frac{n^{n+1}}{n^n + 1}$

- não limitada
- divergente

$w_n = u_n \cdot v_n$

- divergente?
- limitada
- $-1 < w_n < 1$

$$w_n = \frac{n^{n+1}}{n^n+2} \cdot \frac{(-1)^{n+1}}{n} =$$

$$= \frac{n^{n+2} \cdot (-1)^{n+1}}{n(n^n+1)} = \frac{(-n)^{n+1}}{n^{n+1}+n} \Rightarrow n^{n+1}+n > (-n)^{n+1}$$

2. $u_n = \cos(n!\pi)$

- limitada $[-1, 1]$

para $n \geq 2$, $n!$ par

$\Rightarrow \cos(n!\pi) = 1$

logo, $\lim u_n = 1$

1 k par
-1 k impar

$v_n = \frac{n\cos(n\pi)}{2n+1}$

$\frac{n}{2n+1} \cdot \cos(n\pi)$ → 1 ou -1

↳ $(-1)^n$

$\frac{1}{2+0} = \frac{1}{2}$

$v_{2n} \to 1/2$ $v_{n+1} \to -\frac{1}{2}$

não é limite

$a \in \mathbb{R}$

$w_n = \frac{1+a^n}{1+a^{2n}}$

$a > 1$ $a^n \to +\infty$

$\frac{\frac{1}{a^{2n}} + \frac{1}{a^n}}{\frac{1}{a^{2n}} + 1} \to 0$

" "

$a < -1$

$\frac{1+(-1)^n|a|^n}{1+(-1)^{2n}|a|^{2n}} =$

$= \frac{1+(-1)^n|a|^n}{1+|a|^{2n}} =$

$= \frac{1/|a|^{2n} + (-1)^n/|a|^n}{\frac{1}{|a|^{2n}} + 1} \to 0$

$a = 1 \to 1$ $a = -1$ NE lim

3.

a) $\lim \frac{1}{(-1)^n n^2 + 2} = 0$

↳ não é limitada

$u_{2n} =$ $u_{2n+1} =$

b) $\lim \left(1+(-1)^n\right)\left(1+\frac{1}{n}\right)$ → 1 → $u_{2n}=$; → $u_{2n+1}=$

↳ não é convergente

c) $\lim \frac{n\left(1+(-1)^n\right)}{2}$ não é limitada, não é convergente

$u_{2n}=$; $u_{2n+1}=$

d) $\lim \frac{2n^2+(-1)^n}{n^2-1}$ → $u_{2n}=$; $u_{2n+1}=$

e) $\lim \frac{n+\cos n}{2n-1} = \lim \frac{1+\cos n/n}{2-\frac{1}{n}} = \frac{1+0}{2-0} = \frac{1}{2}$

$-\frac{1}{n} \leqslant \frac{\cos n}{n} \leqslant \frac{1}{n}$ (0 ↑ e 0 ↑) por teorema dos sucessões enquadradas

$\frac{\cos n}{n}$ → limitada × $\frac{1}{n}$ → 0, então = 0

f) $\lim \left(-1-\frac{1}{n}\right)^n = -e$

4. u_n convergente

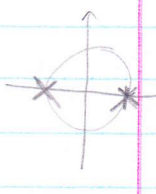
$u_{2n} \in]0,1[$ $u_{2n+1} \in \mathbb{R}\setminus\{0,1\}$ PROVAR: $\lim u_n \in \{0,1\}$

5. $\begin{cases} u_1 = a \\ u_{n+1} = (-1)^n u_n + \frac{u_n}{n+1} \end{cases}$ Se u_n convergente $\lim u_n = 0$

$\overline{\mathbb{R}} = \mathbb{R} \cup]-\infty, +\infty[$

6.

a) $u_n = \frac{1}{n} + 2\cos n\pi = \frac{1}{n} + 2(-1)^n$



$\cos(n\pi) = (-1)^n$ $\quad u_{2n} = \frac{1}{2n} + 2 \to 2 \quad u_{2n+1} = \frac{1}{2n+1} - 2 \to -2$

$S: \{-2, 2\}$

b) $u_{2n} = 0$ e $u_{2n+1} = n$

$S = \{0, +\infty\}$

c) 1, 2, 3, 4, 5, 6, 7, 8 ...

$S = \{+\infty\}$

d) 1, 1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5 ...

$S = \{\mathbb{N}, +\infty\}$

- Sim, como no caso da alínea b
- em $\overline{\mathbb{R}}$ o conjunto dos seublimites nunca é singular se a sucessão for divergente

7. $A = \{x \in \mathbb{R} : |2x+1| < |x|\}$, $B = \{\frac{(-1)^n}{n} : n \in \mathbb{N}_1\}$,

$C = [-1, +\infty[$

a) $|2x+1| < |x| \Leftrightarrow 2x+1 < |x| \vee -(2x+1) < |x| \Leftrightarrow$

$\Leftrightarrow 2x+1 < x \vee 2x+2 < -x \vee -2x-1 < x \vee -2x-1 < -x \Leftrightarrow$

$\Leftrightarrow 2x-x < -1 \vee 2x+x < -1 \vee -2x-x < 1 \vee -2x+x < 1 \Leftrightarrow$

$\Leftrightarrow x < -1 \vee 3x < -1 \vee -3x < 1 \vee -x < 1 \Leftrightarrow$

$\Leftrightarrow x < -1 \vee x < -\frac{1}{3}$

$A =]-1, -\frac{1}{3}[$

b)

$\inf C = -1$ $\quad B = [-1, \frac{1}{2}]$

$\min(C \setminus A) = -1$

$\sup(A \setminus \mathbb{Q}) = -\frac{1}{3}$

$\max(A \cup B) = 1/2$

$\inf B = -1$

$\max(B \setminus \mathbb{Q}) =$ não existe

c) i) sim. Toda a sucessão decrescente e limitada é convergente

ii)

iii) não. Pode convergir para um numero em C

iv) sim. S_n é limitada, pelo teorema de BW garante que u_n tem alguma SS convergente, portanto u_n tem pelo menos um seu limite.

v) sim

8.

a) $A = \{x \in \mathbb{R} : \frac{x^2+1}{x-2} \geq x\}$

$\frac{x^2+1}{x-2} - x \geq 0 \Leftrightarrow$

$\Leftrightarrow \frac{x^2+1-x(x-2)}{x-2} \geq 0 \Leftrightarrow \frac{x^2+1-x^2+2x}{x-2} \geq 0 \Leftrightarrow$

$\Leftrightarrow \frac{2x+1}{x-2} \geq 0$

$x-2 \neq 0 \Leftrightarrow x \neq 2$

$2x+1=0 \Leftrightarrow 2x=-1 \Leftrightarrow x=-\frac{1}{2}$ $\quad A =]-\infty, -\frac{1}{2}] \cup]2, +\infty[$

x	$-\infty$	$-1/2$		2		$+\infty$
$2x+1$	−	0	+	+	+	+
$x-2$	−	−	−	0	+	+
Q	+	0	−	ND	+	+

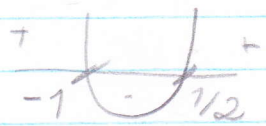
$B = \{x \in \mathbb{R} : \log(2x^2 + x) \geq 0\}$

$\log(2x^2+x) \geq 0 \Leftrightarrow 2x^2 + x \geq 1 \Leftrightarrow 2x^2 + x - 1 \geq 0$

$2x^2 + x - 1 = 0 \Leftrightarrow x = \frac{-1 \pm \sqrt{1 - 4 \cdot 2 \cdot (-1)}}{2 \cdot 2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-1 \pm \sqrt{1+8}}{2 \cdot 2} \Leftrightarrow x = \frac{-1 \pm 3}{4} \Leftrightarrow x = \frac{-1+3}{4} \vee x = \frac{-1-3}{4}$

$\Leftrightarrow x = \frac{1}{2} \vee x = -1$ //



$B =]-\infty, -1] \cup [1/2, +\infty[$

b)

$\min A = NE$

$\sup B \cap \mathbb{Q} = NE$

$\inf A \cap B = NE$

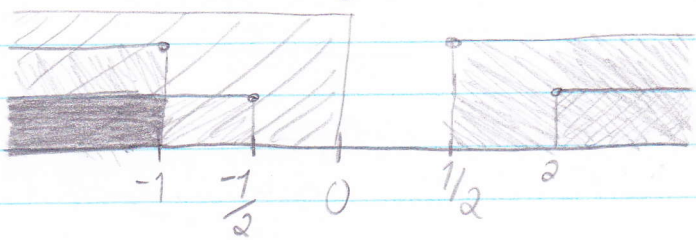
$\sup B \cap \mathbb{R}^- \setminus \mathbb{Q} = -1$

$\max A \cap \mathbb{Q}^- = -1/2$

c) Se u_n é crescente em $A \cap B \cap \mathbb{R}^-$ $]-\infty, -1]$

então é convergente

tende para -1



d) Se u_n é sucessão em $B \cap \mathbb{R}+$ → $[1/2, +\infty[$

então $y_n = (-1)^n x_n$ é divergente

→ $-x_n, x_{n+1}, -x_{n+2}, x_{n+3} \ldots$

$y_{2n} = x_{2n}$

$y_{2n+1} = -x_{2n+1}$

e) $x_n =$

Tender para um nº em $]-1/2, 2]$

Em $A \cap (\mathbb{R} \setminus \mathbb{Q})$ $\quad]-\infty, -\frac{1}{2}] \cup]2, +\infty[$

9) $A = \mathbb{R}^+ \setminus \{1\}$ $\quad B = \{-\frac{1}{n}, n \in \mathbb{N}_1\}$

a) false

b) $A \cap V_{1/2}(0)$

$B^+ \setminus \{1\} \cap]-1/2, 1/2[=]-1/2, 1[\cup]1, 1/2[$

c)

10.

i) $A = [-1+\sqrt{2}, 3]$ $\quad C = \{\frac{1}{k^2}, k \in \mathbb{N}_1$

ii) $]-\infty, -1+\sqrt{2}[\cup]3, +\infty[$ $\quad a_{n+2} a_n < 0$

iii) Verdadeiro $\quad a_n \to 0$

11.

a) $1 - \sqrt{n} \to -\infty$

$\lim \sqrt{n} = \lim \frac{n+1}{n} = \lim \frac{n/n + 1/n}{n/n}$

$= \lim 1 + \frac{1}{n} = 1 + 0 = 1$

$\lim 1 - \sqrt{n} = 1 - 1 = 0$

?

b) $\frac{n^2+1}{n} \to +\infty$

$\lim \frac{n^2+1}{n} = \lim \frac{n^2/n^2 + 1/n^2}{n/n^2} =$

$= \frac{1+0}{0} =$

?

$n^p < a^n < n! < n^n$

12.

✓ a) $\lim \frac{n^n}{7000^n} = +\infty$

✓ b) $\lim n^{n+1} - n^n = \lim n^n(n-1) = +\infty(+\infty - 1) = +\infty$

ou

$n^{n+1}\left(1 - \frac{1}{n}\right) = +\infty$

c) $\lim 3^n - (2n)! = -\infty$

✓ d) $\lim (n! - n^{7000})^n = \left[n!\left(1 - \frac{n^{7000}}{n!}\right)\right]^n \geq n!\left(1 - \frac{n^{7000}}{n!}\right)$

$\to 0$ // $\to +\infty$

e) $\lim \frac{(2n)!}{n!} = \frac{+\infty}{+\infty} = +\infty$

f) $\lim \sqrt[n]{\frac{n+2}{n+2}} = 1$

✓

$\lim \frac{a_{n+1}}{a_n} = \ell$?

$\lim \frac{n+3}{n+2} \cdot \frac{n+1}{n+2} = 1 \cdot 1 = 1$

✓ g) $\lim \frac{n^{7000}}{1,0002^n} = 0$ $\qquad \frac{n^p}{a^n} = 0$

h) $\lim \sqrt[n]{\frac{n}{n^2+1}} = 1$

$\lim \frac{\frac{n+1}{(n+1)^2+1}}{\frac{n}{n^2+1}} = \lim \frac{n+1}{n} \cdot \frac{n+1}{n^2+2n+2} = \lim 1 + \frac{1}{n} \cdot \frac{0+1}{1+0+0}$

$= 1 + 0 \cdot \frac{1}{1} = 1$

$\lim \left(1+\frac{1}{n}\right)^n = e$ $\quad \left(1+\frac{a}{n}\right)^n \to e^a$

$\left(1-\frac{1}{n}\right)^n = e^{-1}$

i) $\sqrt[n]{3^n+2} \to 3$

$$\lim \frac{3^{n+1}+2}{3^n+2} = \lim \frac{1+2/3^{n+1}}{\frac{1}{3}+2/3^{n+1}} = \frac{1+0}{\frac{1}{3}+0} = 3$$

j) $\lim \sqrt[n]{n!} = +\infty$

$$\lim \frac{(n+2)!}{n!} = \lim \frac{n!(n+1)}{n!} = \lim n+1 = +\infty$$

k) $\lim \left(2-\frac{1}{n}\right)^n \geq \lim \left(\frac{3}{2}\right)^n = +\infty$ $\quad 2-\frac{1}{n} \to 2$ $\quad \exists p \in \mathbb{N}\ \forall n > p$ $\quad 2-1/n \geq \frac{3}{2}$ → exemplo entre 1 e 2

l) $\lim \left(1-\frac{1}{2^{n-1}}\right)^{2^n} = \lim \left(\left(1-\frac{1}{2^{n-1}}\right)^{2^{n-1}}\right)^2 = \lim \left(\left(1-\frac{1}{2^{n-1}}\right)^{2^{n-1}}\right)^2 =$

$= e^{-2}$

m) $\lim \left(1+\frac{1}{n}\right)^{n^2} = \lim \left(\left(1+\frac{1}{n}\right)^n\right)^n = e^n = +\infty$

13.

a) $\lim \frac{2n+3}{3n-1} = \frac{2}{3}$

b) $\lim \frac{n^2+1}{n^4+3} = \lim \frac{n^2/n^4 + 1/n^4}{1+3/n^4} = \frac{0+0}{1+0} = 0$

c) $\lim \frac{2^n+1}{2^{n+1}-1} = \lim \frac{2^n/2^{n+1} + 1/2^{n+1}}{1-1/2^{n+1}} = \frac{0+0}{1-0} = 0$

d) $\lim \frac{n^3+1}{n^2+2n-1} = \lim \frac{n^2\left(n+\frac{1}{n^2}\right)}{n^2+2n-1} = \frac{1.(0+0)}{1+0+0} = 0$

e) $\lim \frac{(-1)^n n^3 + 1}{n^2 + 2} =$

$=$

f) $\lim \frac{n^p}{n!} = +\infty$

g) $\sqrt[n]{1 + \frac{1}{n}} = 1$ ou $\left(1 + \frac{1}{n}\right)^{1/n} = 1^0 = 1$

$\frac{a_{n+1}}{a_n} = \lim \frac{(n+1)+1}{n+1} \cdot \frac{n}{1+n} = \lim \frac{n(n+2)}{n+1} =$

$= \frac{1(1+0)}{1+0} = 1$

h) $\lim \frac{\left(\frac{1}{2}\right)^n}{n^3} = +\infty$

i) $\lim \frac{3^n}{n^2} = +\infty$

j) $\sqrt[n]{\frac{n^2 + n - 1}{n+3}}$

$\lim \frac{a_{n+1}}{a_n} = \lim \frac{(n+1)^2 + n}{n+4} \cdot \frac{n+3}{n^2+n-1} = \lim \frac{n^2+3n+1}{n+4} \cdot \frac{n+3}{n^2+n-1} =$

$= +\infty \cdot \frac{0+0}{1+0-0}$

k) $\sqrt[n]{2^n+1}$

$$\lim \frac{2^{n+1}+1}{2^n+1} = +\infty$$

l) $\sqrt[n]{(n+1)!-n!} \to +\infty$

$$\frac{a_{n+1}}{a_n} \quad \frac{(n+2)!-(n+1)!}{(n+1)!-n!} = \frac{(n+1)!\cdot(n+2)-(n+1)!}{n!\cdot(n+1)!-n!} =$$

$$\lim \frac{1\cdot(n+2)-1}{1-n!/(n+1)!} = \frac{n+1}{1-0} = +\infty$$

m) $\lim \left(1+\frac{1}{n^2}\right)^{n^3} = \lim \left(\left(1+\frac{1}{n^2}\right)^{n^2}\right)^n = e^n$

n) $\lim \left(1-\frac{1}{n!}\right)^{n!} = e^{-1}$

a) $\lim \left(1+\frac{1}{n^3}\right)^{n^2} = \lim \left(\left(1+\frac{1}{n^3}\right)^{n^3}\right)^{\frac{1}{n}} = e^{\frac{1}{n}}$

14.

a) $\lim \frac{n!}{n^{7000}} = +\infty$

b) $\lim \frac{(2n)!+2}{(3n)!+3} = \lim \frac{2+2/n!}{3+3/n!} = \frac{2}{3}$

c) $\lim \frac{(2n)!}{(2n)^n} = 0$

d) $\lim \frac{(n!)^2}{(2n)! + 2} =$

e) $\lim \frac{2^n \cdot n!}{n^n} = 0$

f) $\lim \frac{3^n n!}{n^n} = 0$

g) $\lim n^{1/n} = \sqrt{n} = 1$

h) $\lim \left(\frac{1}{n}\right)^{2/n} = \lim \sqrt[n]{1/n} = 0$

$\lim \frac{1}{n+1} \cdot \frac{1}{n} = \lim \frac{1}{n^2+n} = 0$

i) $\lim \left(\frac{1}{n}\right)^n = \lim \frac{1}{n^n} = 0$

j) $\lim \left(2 - \frac{1}{n}\right)^n = 2^{+\infty} = +\infty$

k) $\lim \left(\frac{n-1}{2n^2+1}\right)^{2/n} = \left(\lim \sqrt[n]{\frac{n-1}{2n^2+1}}\right)^2 = 1$

$\lim \frac{n}{2n^2+4n} \cdot \frac{2n^2+1}{n-1} = \lim \frac{n}{n-2} \cdot \frac{2n^2+1}{2n^2+4n} =$

$= 1 \cdot 1 = 1$

f) $\lim \frac{2^{(n^2)}}{75^n} = \lim \left(\frac{2^n}{75}\right)^n \geq \frac{2^n}{75}$

$\hookrightarrow +\infty$

15.

a) PROVÁMOS TEORICA

i) se $u_n \to +\infty$ NP então $\frac{1}{u_n} \to 0$

ii) se $u_n > 0$ e $u_n \to 0$ então $\frac{1}{u_n} \to +\infty$

b) $u_n \to 0 \overset{?}{\Rightarrow} \left(\frac{1}{u_n} \to +\infty \vee \frac{1}{u_n} \to -\infty\right)$

SSe $u_n > 0$

Não é verdade

$u_n = -\frac{1}{n}$

1

Ficha 6

1. a) $y = e^{x^2-2}$ $\quad x>0$

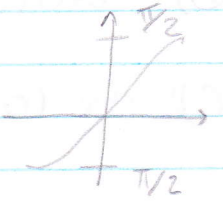
$\Leftrightarrow \log y = x^2 - 2 \Leftrightarrow \log y + 2 = x^2 \Leftrightarrow$

$\Leftrightarrow \sqrt{\log y + 2} = F^{-1}(y)$

$D' =]e^{-2}, +\infty[$

$D =]0, +\infty[$

b) $2\operatorname{sen} x$, $x \in]-\pi/2, \pi/2[$

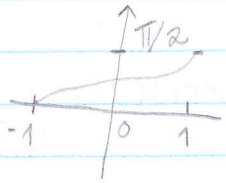


$D' =]-\pi/2, \pi/2[$ $\quad 2\operatorname{sen} x = y \Leftrightarrow$

$D =]-1, 1[$ $\quad \Leftrightarrow \operatorname{sen} x = y/2 \Leftrightarrow$

$\Leftrightarrow \dfrac{\text{arcoseno } x}{2}$

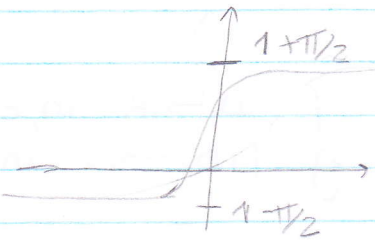
c) $\cos(2x)$ $\quad x \in]0, \pi/2[$



$D' =]0, \pi/2[$

$D =]-1, 1[$ $\quad$ arcocoseno $\left(\frac{1}{x}\right)$ ou arcocoseco$(2x)$?

d) $\operatorname{tg}(x-1)$ $\quad x \in]1-\pi/2, 1+\pi/2[$



$D' =]1-\pi/2, 1+\pi/2[$

$D = \mathbb{R}$

arcotangente$(x-1)$

2. $\arccos 0 = k\pi + \dfrac{\pi}{2}$ $\quad \arccos 1 = 2k\pi$

$\arccos x = y \Rightarrow \cos y = x$ $\quad \arccos\left(-\frac{1}{2}\right) = \frac{4\pi}{3}$ e

arcsen$\left(\frac{-1}{2}\right)$

3. $\arcsen a = x \iff \sen x = a$
$\arccos a = x \iff \cos x = a$
$\arctg a = x \iff \tg x = a$

4.

a) $\cos(\arccos x) = y \iff \arccos y = \arccos x$

b) $\sen(\arcsen x) = x \iff \arcsen x = \arcsen x$

c) $\cos(\arcsen x) = \sqrt{1-x^2} \iff \arccos(\sqrt{1-x^2}) = \arcsen x$

d) $\sen(\arccos x) = \sqrt{1-x^2} \iff \arcsen(\sqrt{1-x^2}) = \arccos x$

e) $\tg(\arcsen x) = \frac{x}{\sqrt{1-x^2}} \iff \arctg\left(\frac{x}{\sqrt{1-x^2}}\right) = \arcsen x$

f) $\tg(\arccos x) = \frac{\sqrt{1-x^2}}{x} \iff \arctg\left(\frac{\sqrt{1-x^2}}{x}\right) = \arccos x$

5. $f: D \to \mathbb{R}$ injectiva
$g: f(D) \to D$ inversa de f
$\Big|\ g(y) = x$ sse $f(x) = y$

a) f crescente: $\forall x_1, x_2 \in D : x_1 \leq x_2 \iff f(x_1) \leq f(x_2)$

(f decrescente é análoga)

$\Downarrow$ g é inversa f

$f(x_1) \leq f(x_2) \iff x_1 \leq x_2$

$\Downarrow$

$y_1 \leq y_2 \iff g(y_1) \leq g(y_2)$

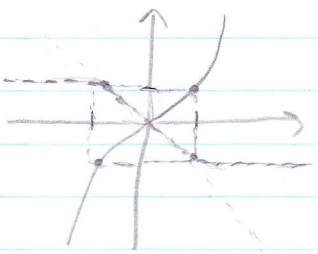
b) f impar: $\forall x \in D$

$f(x) = -f(-x)$

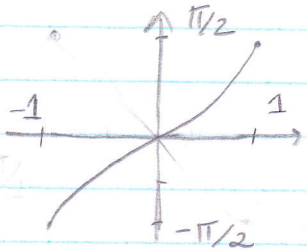
$\Downarrow$ g é inversa de f

$g(f(x)) = -g(f(-x))$

$\iff g(y) = -g(-y)$

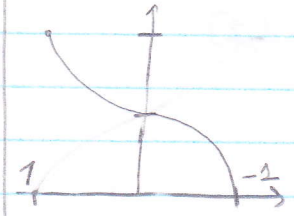


c) arcsen



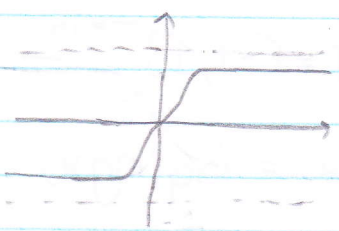
$\forall x_1, x_2 \in [-1,1], x_1 < x_2$
$\arcsen x_1 < \arcsen x_2$
e
$\forall x \in [-1,1]$
$\arcsen x = -\arcsen(-x)$

arccos



$\forall x_1, x_2 \in [-1,1]$
$x_1 < x_2$
$\arccos x_1 > \arccos x_2$
$\forall x \in [-1,1]$
$\arccos x = \arccos(-x)$

arctg



$\forall x_1, x_2 \in \mathbb{R}$
$x_1 < x_2$
$\arctg x_1 < \arctg x_2$

$\forall x \in \mathbb{R}$
$\arctg x = -\arctg(-x)$

6)

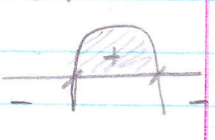
a) $f(x) = \dfrac{x}{\sqrt{4-x^2}}$

$D = \{x : \sqrt{4-x^2} \neq 0 \wedge 4-x^2 \geq 0\}$

$= \{x : -x^2+4 > 0\}$

$=]-2,2[$

$-x^2+4 = 0 \Leftrightarrow$
$\Leftrightarrow 4 = x^2 \Leftrightarrow x = \pm\sqrt{4}$
$\Leftrightarrow x = \pm 2$



b) $f(x) = \dfrac{1}{\cos^2 x} + \dfrac{1}{\sen^2 x}$

$Df = \{x : \cos^2 x \neq 0 \wedge \sen^2 x \neq 0\}$

$= \mathbb{R} \setminus \{k\pi, k \in \mathbb{Z}\} \cup \mathbb{R} \setminus \{k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$

$= \mathbb{R} \setminus \{\frac{k\pi}{2} : k \in \mathbb{Z}\}$

c) $f(x) = \underset{\hookrightarrow \mathbb{R}}{\operatorname{tg} x} + \underset{\hookrightarrow [-1,1]}{\operatorname{cotg} x}$

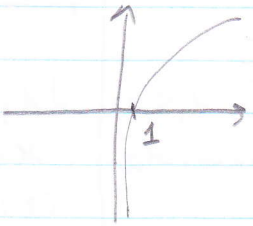
$Df = [-1,1]$

c|) $f(x) = \log(\log x)$

$Df = \{x : \log x > 0 \wedge x > 0\}$

$= \{x : x > 1 \wedge x > 0\}$

$= \{x : x > 1\} =]1, +\infty[$



e) $f(x) = \log(1 - x^{3/2})$

$Df = \{x : 1 - x^{3/2} > 0\}$

$= \{x : x > 1\} =]1, +\infty[$

$1 - x^{3/2} > 0 \iff$

$\iff \sqrt{x^3} > 1 \iff$

$\iff x^3 > 1^2 \iff x > \sqrt[3]{1} \iff x > 1$

$(x^3)^{\frac{1}{2}} = \sqrt{x^3}$

f) $f(x) = \operatorname{arctg}\left(\frac{1}{1-x^2}\right)$

$df(x) = \{x : 1 - x^2 \neq 0\}$

$= \mathbb{R} \setminus \{-1, 1\}$,,

g) $f(x) = \arccos\left(\frac{1}{x}\right)$

$Df(x) = \{x : x \neq 0 \ \text{e} \ -1 \leq \frac{1}{x} \leq 1\}$

$= \{x : x > 0 \wedge x \geq 1 \vee x < 0 \wedge x \leq -1\}$

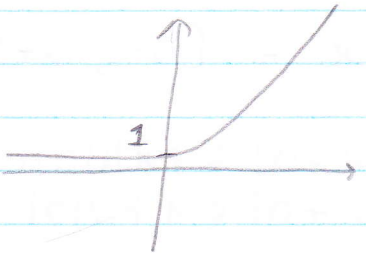
$= \mathbb{R} \setminus]-1, 0, 1[$

h) $f(x) = \arcsen(e^x)$

P.V.

$D = \{x : -1 \leq e^x \leq 1\}$

$= \{x : e^x \leq 1\} =]-\infty, 0]$



i) $f(x) = \log(1 - \arcsen x)$

$D = \{x : 1 - \arcsen x > 0 \wedge -1 \leq x \leq 1\}$

$= \{x : x < \sen 1 \wedge -1 \leq x \leq 1\}$

$= \{x : -1 \leq x < \sen 1\}$

$= [-1, \sen 1[$

$1 - \arcsen x > 0 \Leftrightarrow$
$\Leftrightarrow \arcsen x < 1 \Leftrightarrow$
$\Leftrightarrow x < \sen 1$

7. u_n é monotona $\Rightarrow \operatorname{arctg} u_n$ é convergente

$-\pi/2 < \operatorname{arctg} x < \pi/2$, limitada

a função $\operatorname{arctg} u_n$ é composta por uma função limitada e uma sucessão monotona, logo é convergente

8) $f(x) = x^2 + 1 \qquad g(x) = |x|$

As funções polinomiais e modulo são continuas em todo o seu dominio

$\forall \varepsilon > 0 \ \exists \delta > 0 : |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$

$|1 + x^2| - |1 + a^2| < \varepsilon \Leftrightarrow |x^2 - a^2| < \varepsilon \Leftrightarrow$

$\Leftrightarrow |(x + a)(x - a)| < \varepsilon$ → encontrar um $\delta > 0$

Se $x \in [a-1, a+1] \rightarrow \delta < 1$

$|x+a| \leq |x|+|a| \Leftrightarrow |x+a| \leq 1+|a|+|a| \Leftrightarrow$
$\Leftrightarrow |x+a| \leq 1+2|a|$

Se $|x-a| \cdot (1+2|a|) < \varepsilon \Rightarrow |(x-a)(x+a)| < \varepsilon$

$|(x-a)(x+a)| < |(x-a)(1+2|a|)| < \varepsilon$

basta $\delta = \dfrac{\varepsilon}{1+2|a|}$, porque $|x-a| < \delta \Rightarrow |x-a| \cdot (1-2|a|) <$
$< \delta \cdot (1-2|a|) = \varepsilon$

9. $\lim x_n = 1$, $x_n > 1 \Rightarrow$ Convergente

a) $f(x) = \dfrac{1}{x}$ $x \neq 0$ → limitada

$\lim f(x_n) = \lim \dfrac{1}{x_n} = \dfrac{1}{1} = 1$

b) $f(x) = \log x$, $x > 0$ → não é convergente

$\lim_{x \to +\infty} \log x = +\infty$

$f(x_n) = \log(x_n)$ $x_n > 1$

$\lim f(x_n) = \lim \log(1) = 0$

c) $f(x) = \dfrac{1}{x-1}$, $x \neq 1$

10. $\phi : [a,b] \to \mathbb{R}$ <u>contínua</u> a e $b \in \mathbb{R}$ $a<b$

$\exists\, x_n$, termos em $[a,b]$

$\lim \phi(x_n) = 0$

↳ sublimite de ϕ

Se ϕ é contínua em $[a,b]$ e $\lim \phi(x_n) = 0$ então $x_n = 0$.

⇓

Como $a<b$ $\exists$ ponto c entre $]a,b[$ em que $f(c) = 0$

$$\lim_{x \to a} f(x) = f(a)$$

11. $g : [0,1] \to \mathbb{R}$ contínua

a)

b) se $\exists\, x_n$ em $[0,1]$ tq $g(x_n) = \frac{1}{n}$ $\forall n \in \mathbb{N}$

então $\exists\, c \in [0,1]$ tq $g(c) = 0$

12.

a) $\frac{x+1}{x^3+x} = \frac{x+1}{x(x^2+1)}$ — $x+1$: contínua; $x(x^2+1)$: contínua

$x = 0$?

$\lim_{x \to 0} \frac{x+1}{x(x^2+1)} = \frac{1}{0} = \pm\infty$? Ponto de descontinuidade

b) $\frac{x+1}{x^4+3x^3+2x^2} = \frac{x+1}{x^2(x+2)(x+1)}$ (x+1: contínua; $x^2(x+2)(x+1)$: contínua) $D_f = \mathbb{R} \setminus \{-2, -1, 0\}$

$x^4+3x^3+2x^2 = x^2(x^2+3x+2) = x^2(x+2)(x+1)$

$x^2+3x+2 = 0 \Leftrightarrow x = \frac{-3 \pm \sqrt{9-4.1.2}}{2} \Leftrightarrow x = \frac{-3 \pm 1}{2} \Leftrightarrow$

$\Leftrightarrow x = \frac{-2}{2} \vee x = \frac{-4}{2} \Leftrightarrow x = -1 \vee x = -2$

$\lim_{x \to 2^-} \frac{x+1}{x^2(x+2)(x+1)} = -\infty$ ($x^2(x+2)(x+1) \to 0^-$)

Descontínua em $x = -2$

$\lim_{x \to -2^+} \frac{x+1}{x^2(x+2)(x+1)} = +\infty$ ($x^2(x+2)(x+1) \to 0^+$)

$\lim_{x \to 0} \frac{x+1}{x^2(x+2)(x+1)} = \frac{1}{0.2.1} = \frac{1}{0} = +\infty$ Descontínua em $x = 0$

$\lim_{x \to -1} \frac{\cancel{x+1}}{x^2(x+2)\cancel{(x+1)}} = \lim_{x \to -1} \frac{1}{x^2(x+2)} = \frac{1}{1.1} = 1$

Prolongado a 1

c) $\sqrt{x} - \frac{1}{x^2+x} = \sqrt{x} - \frac{1}{x(x+1)}$ ($x(x+1)$: contínua; $\sqrt{x}$: contínua ($x \geq 0$))

$\lim_{x \to 0^+} \sqrt{x} - \frac{1}{x(x+1)} =$

$= 0^+ - \frac{1}{0} = -\infty$

d) $\operatorname{sen}(\cos\sqrt{1-x^2})$ → contínua em todo o domínio

$D = \{x : 1-x^2 \geq 0\}$

$= \{x : -1 \leq x \leq 1\}$

$= [-1, 1]$

$\lim_{x \to -1} \operatorname{sen}(\cos\sqrt{1-x^2}) = \operatorname{sen}(\cos 0) = \operatorname{sen} 1$

prolongado a sen1

$\lim_{x \to 1} \operatorname{sen}(\cos\sqrt{1-x^2}) = \operatorname{sen} 1$

e) $\cos\frac{1}{\sqrt{1-x^2}}$ → contínua em todo o domínio

$D = \{x : \sqrt{1-x^2} \neq 0 \wedge 1-x^2 \geq 0\}$

$= \{x : -1 < x < 1\}$

$\lim_{x \to 1^+} \frac{1}{\sqrt{1-1^2}} = \frac{1}{0} = +\infty = \lim_{x \to 1} \frac{1}{\sqrt{1-x^2}}$

descontínua em 1 e -1

f) $\sqrt[3]{\operatorname{tg} 2x - \operatorname{cotg} 2x}$

P descontinuidade → $\lim_{x} f(x)$ e $\lim_{x \pm} f(x)$

g) $\frac{1}{\sqrt{x^2+1}} + \frac{1}{\sqrt[3]{x^3-1}}$

$D = \{x : \sqrt{x^2+1} \neq 0 \wedge x^2+1 \geq 0 \wedge \sqrt[3]{x^3-1} \neq 0\}$

$= \{x : x \neq 1\}$

$= \mathbb{R} \setminus \{1\}$

$x^3 - 1 = 0 \Leftrightarrow$

$\Leftrightarrow x^3 = 1 \Leftrightarrow x = 1$

$\lim_{x \to 1} f(x) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt[3]{1^2-1}} = \frac{1}{2} + \infty = +\infty$

descontínuo em 0

h) $\frac{|x^2-1|}{x^\infty - 1}$

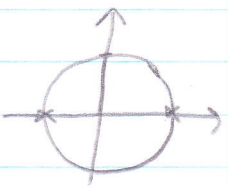
$D = \{x : x^2 - 1 \neq 0\}$

$= \{x \neq 1\} \quad = \mathbb{R} \setminus \{1\}$

$\lim_{x \to 1^-} \frac{|x^2-1|}{x^2-1} = -\infty$ (0^-)

$\lim_{x \to 1^+} \frac{|x^2-1|}{x^2-1} = +\infty$ (0^+, 0^-, 0^+)

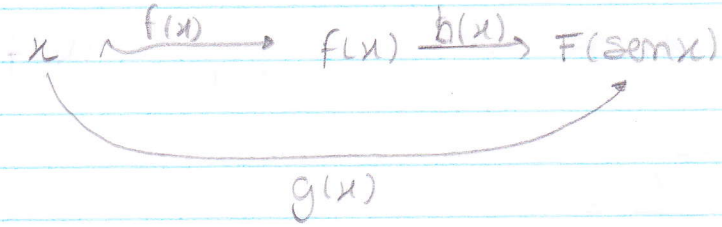
i) $\sqrt{-\operatorname{sen}^2 x}$



$D = \{x : -\operatorname{sen}^2 x \geq 0\}$ $\quad -\operatorname{sen}^2 x = 0$

13. $f: \mathbb{R} \to \mathbb{R}$ contínua em $x=1$

onde é contínua $g(x) = f(\operatorname{sen} x)$



- $\operatorname{sen} x$ é contínua em toda o domínio ↳ $\mathbb{R}$
- contra domínio = $[-1, 1]$
- se $f(x)$ é contínua em $x=1$, $g(x)$ é contínua quando $\operatorname{sen} x = 1 \Leftrightarrow x = \frac{\pi}{2} + 2k\pi$

14. $f: \mathbb{R} \to \mathbb{R}$ contínua em $x=0$

$$g(x) = f(\operatorname{tg} x - \operatorname{cotg} x)$$

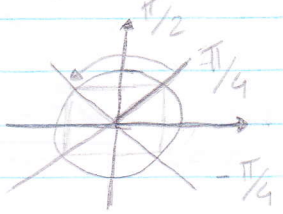
$g(x)$ é obrigatoriamente contínua quando $(\operatorname{tg} x - \operatorname{cotg} x) = 0$

$$\operatorname{tg} x - \operatorname{cotg} x = 0 \Leftrightarrow \operatorname{cotg} x = \operatorname{tg} x \Leftrightarrow \frac{1}{\operatorname{tg} x} = \operatorname{tg} x \Leftrightarrow$$

$$\Leftrightarrow 1 = \operatorname{tg} x \cdot \operatorname{tg} x \Leftrightarrow 1 = (\operatorname{tg} x)^2 \Leftrightarrow \sqrt{1} = \operatorname{tg} x \Leftrightarrow$$

$$\Leftrightarrow \operatorname{tg} x = \pm 1 \Leftrightarrow x = \frac{\pi}{4} + \frac{k\pi}{2} \Leftrightarrow$$

$$\Leftrightarrow x = \frac{\pi + 2k\pi}{4}$$



15. $f: \mathbb{R} \to \mathbb{R}$ $f(x) = x\,d(x)$

$d: \mathbb{R} \to \mathbb{R}$ dirichelet

$$d(x) = \begin{cases} 1 \text{ se } x \in \mathbb{Q} \\ 0 \text{ se } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

// descontínua

$$f(x) \begin{cases} 0 \text{ se } x \in \mathbb{R} \setminus \mathbb{Q} \\ x \text{ se } x \in \mathbb{Q} \end{cases}$$

$$\lim_{x \to 0} f(x) = 0 = f(0)$$

16.

a) $\lim_{x \to 0} \frac{1}{x^2} = +\infty$

$\forall \varepsilon > 0 \ \exists R > 0 \ \forall x \in D \quad 0 < |x| < R \Rightarrow F(x) > \varepsilon$

$\frac{1}{x^2} > \varepsilon \quad 0 < |x| < R$

$\Leftrightarrow 1 > \varepsilon x^2 \Leftrightarrow \sqrt{\frac{1}{\varepsilon}} > x$

b) $\lim_{x \to +\infty} \frac{1}{x} = 0$

$\forall \varepsilon > 0 \ \exists R > 0 \ \forall x \in D \quad x > R \Rightarrow F(x) < \varepsilon$

c) $\lim_{x \to +\infty} \sqrt{x} = +\infty$

$\forall \varepsilon > 0 \ \exists R > 0 \ \forall x \in D \quad x > R \Rightarrow f(x) > \varepsilon$

17.

a) $\lim_{x \to 0} \frac{x^3 - x^2 + x - 1}{x^2 - 1} = \frac{-1}{-1} = 1$

b) $\lim_{x \to 1} \frac{x^3 - x^2 + x - 1}{x^2 - 1} = \lim_{x \to 1} \frac{(x-1)(x^2+1)}{(x+1)(x-1)} = \frac{2}{2} = 1$

$x^3 - x^2 + x - 1 \ | \ x - 1$

$x^3 - x^2 \qquad\qquad x^2 + 1$

$0 \quad -x - 1$

0

c) $\lim_{x\to 0} \frac{e^{x^2}-1}{x} = \lim_{x\to 0} x\left(\frac{e^{x^2}-1}{x^2}\right) = \overset{y=x^2}{}$

$= \lim_{x\to 0} x \cdot \lim_{x\to 0} \frac{e^y-1}{y} = 0.1$

d) $\lim_{x\to 0} \left[x^2\left(1-\cos\frac{1}{x}\right)\right]$

e) $\lim_{x\to 0} \operatorname{sen}\frac{1}{x}$

$\lim_{x\to 0^+} \operatorname{sen}(1/x) = \lim_{y\to+\infty} \operatorname{sen} y$

pelo critério de Heine:

$y_k = 2\pi k \to +\infty \quad \operatorname{sen} y \to 0$
$y_k = 2\pi k + \frac{\pi}{2} \to +\infty \quad \operatorname{sen} y \to 1$

a função tem dois sublimites diferentes,
logo $\lim_{x\to 0} \operatorname{sen}\frac{1}{x}$ não existe

f) $\lim_{x\to+\infty} \operatorname{sen}\underbrace{\frac{1}{x}}_{\to 0} = \operatorname{sen} 0 = 1$

g) $\lim_{x\to 0} x \operatorname{sen}\frac{1}{x} = 0$

Composta por x (tende p/ 0) e $\operatorname{sen}\frac{1}{x}$ (limitada)

18.

a) $\lim_{x \to 1} \frac{x^2 - x}{x^2 - 3x + 2} = \lim_{x \to 1} \frac{x(x-1)}{(x-1)(x-2)} =$

$= \frac{1}{1-2} = -1$

b)

$x = \frac{3 \pm \sqrt{9 - 4 \cdot 1 \cdot 2}}{2} \Leftrightarrow x = \frac{3 \pm 1}{2} \Leftrightarrow$

$\Leftrightarrow x = 2 \vee x = 1$

$\lim_{x \to 0} \frac{\operatorname{tg} 5x}{x \arccos x} =$

Laura : 2-N 4.18

15:30h - 18h

Ficha 7

1. $f\left(-\frac{1}{n}\right) = 1 - f\left(\frac{1}{n}\right)$

$$f(0^-) + f(0^+) = \lim_{x\to 0^-} f(x) + \lim_{x\to 0^+} f(x)$$

Pelo critério de Heine:

$$\frac{1}{n} \to 0^+ \Rightarrow f\left(\frac{1}{n}\right) \to f(0^+)$$

$$-\frac{1}{n} \to 0^- \Rightarrow f\left(-\frac{1}{n}\right) \to f(0^-)$$

$$\lim_{x\to 0^-} f\left(-\frac{1}{n}\right) = \lim_{x\to 0^-} 1 - f\left(\frac{1}{n}\right) \Leftrightarrow f(0^-) = 1 - f(0^+) \Leftrightarrow$$

$$\Leftrightarrow f(0^-) + f(0^+) = 1 \quad //$$

---

se $\exists \lim_{x\to 0} f(x) = a$

então $f(0^-) = f(0^+) = a$ e $a + a = 1 \Leftrightarrow$

$$\Leftrightarrow a = \frac{1}{2} \quad //$$

2. $f(x) = \frac{x + |x|}{2}\, d(x)$ → dirichelet

a) $d(x) \begin{cases} 0 & \text{se } x \notin \mathbb{Q} \\ 1 & \text{se } x \in \mathbb{Q} \end{cases}$ minorada, min = 0

$$f(x) = \begin{cases} 0 & \text{se } x \notin \mathbb{Q} \wedge x \leq 0 \\ x & \text{se } x \in \mathbb{Q} \end{cases}$$

$\mathbb{Q} = \mathbb{Q}_0^+$

b) $\lim_{x \to -\infty} f(x) = 0$

$\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} x = +\infty$

c) Em que pontos f é contínua?

em $]-\infty, 0]$ $\lim_{x \to a} f(x) = f(a)$ $a \in]-\infty, 0]$

3. $f: \mathbb{R} \to \mathbb{R}$ contínua em $x = 1$

$$f(x) = \begin{cases} 0 & , \text{se } x \leq -1 \\ \arcsen x & , \text{se } -1 < x < 1 \\ k \operatorname{sen}\left(\frac{\pi}{2} x\right) & , \text{se } x \geq 1 \end{cases}$$

a)

$\lim_{x \to 1^+} f(x) = f(1) = k \operatorname{sen}\left(\frac{\pi}{2} \cdot 1\right) = k$

$k = \lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} \arcsen x = \arcsen 1 = \frac{1}{2}$

b) → Para $x \in \mathbb{R} \setminus \{-1, 1\}$ f é contínua, pois é dada por funções contínuas.

→ f é contínua em $x = 1$

f é contínua em $x = -1$?

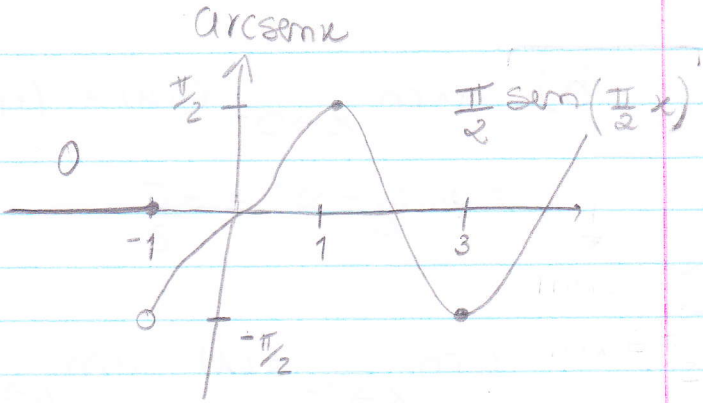
$f(-1) = 0$ $\lim_{x \to -1^-} f(x) = 0 \neq$

não é contínua em $x = -1$

$\lim_{x \to -1^+} f(x) = \arcsen(-1) = -\pi/2$

c)

- não tem supremo
- não tem máximo

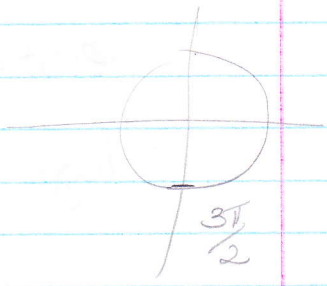


$\underbrace{\frac{\pi}{2}\operatorname{sen}\left(\frac{\pi}{2}x\right)}_{[-1,1]}$ $\begin{cases} max = \pi/2 \\ min = -\pi/2 \end{cases}$

$\frac{\pi}{2}\operatorname{sen}\left(\frac{\pi}{2}x\right) = -\frac{\pi}{2} \Leftrightarrow$

$\Leftrightarrow \frac{\pi}{2}\cdot(-1) = -\frac{\pi}{2} \Leftrightarrow \frac{\pi}{2}x = \frac{3\pi}{2} \Leftrightarrow$

$\Leftrightarrow x = 3$



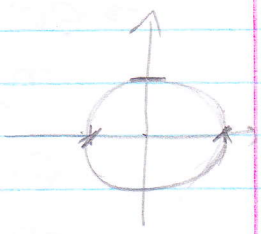
f é contínua em $[1,3]$, $f(1) = \pi/2$ e $f(3) = -\pi/2$
Pelo teorema do valor intermédio, existe pelo menos um $c \in [1,3]$ em que $f(c) = 0$

d) $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} 0 = 0$

$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} \frac{\pi}{2}\operatorname{sen}\left(\frac{\pi}{2}x\right)$

$u_k = \frac{\pi}{2} + 2k\pi \to 1$
$\neq$
$u_k = \pi k \to 0$

logo $\nexists \lim_{x\to+\infty} f(x)$



4. $\varphi : \mathbb{R} \to \mathbb{R}$ $\begin{cases} \operatorname{arctg}\frac{1}{x}, & \text{se } x < 0 \\ 1 + e^{1-x}, & \text{se } x \geq 0 \end{cases}$

a) - a função $\operatorname{arctg} x$ é contínua em $\mathbb{R}$, logo $\operatorname{arctg}\frac{1}{x}$ é contínua em $\mathbb{R}\setminus\{0\}$

- $1 + e^{1-x}$ é contínua, porque é composta por uma constante e uma função exponencial (ambas contínuas)

b) $\lim_{x\to 0^-} e(x) = \lim_{x\to 0^-} \operatorname{arctg}\frac{1}{x} =$

$\frac{1}{x} \to -\infty$

$= \operatorname{arctg}(-\infty) = -\frac{\pi}{2}$

$\lim_{x\to 0^+} e(x) = \lim_{x\to 0^+} 1 + e^{1-x} = 1 + e$

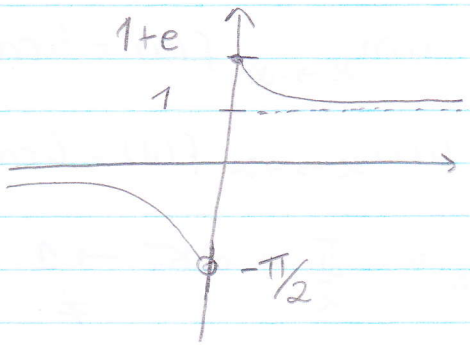
$e(0) = 1+e = \lim_{x\to 0^+} e(x) \neq \lim_{x\to 0^-} e(x)$

logo, $e(x)$ é contínua à direita no ponto $x=0$
$e(x)$ não é contínua à esq no ponto $x=0$,
não é contínua em $x=0$

c) $\lim_{x\to+\infty} e(x) = \lim_{x\to+\infty} 1 + e^{1-x} = 1 + e^{-\infty} = 1 + 0 = 1$

$\lim_{x\to-\infty} e(x) = \lim_{x\to-\infty} \operatorname{arctg}\frac{1}{x} = \operatorname{arctg}\frac{1}{+\infty} = \operatorname{arctg} 0 = 0$

d) $D' =]-\pi/2, 0[\cup]1, 1+e]$



Pelo T do valor intermédio

→ $e(0) = 1+e$

$\lim_{x\to+\infty} e(x) = 1$

$e(x)$ atinge todos os pontos entre $]1, 1+e]$

→ $\lim_{x\to 0^-} e(x) = -\pi/2$

$\lim_{x\to-\infty} e(x) = 0$

$e(x)$ atinge todos os pontos em $]-\pi/2, 0[$

$x \geq 0$

$0 < e^{-x} \leq 1 \Leftrightarrow 0 < e^{1-x} \leq e \Leftrightarrow 1 < 1 + e^{1-x} \leq e+1$

$x<0$

$$\frac{1}{x}<0 \iff -\pi/2 < \operatorname{arctg}\frac{1}{x}<0$$

5.

a) $\varphi(x) = e^{-\frac{1}{x^2}}$

é contínua em $\mathbb{R}\setminus\{0\}$ porque é composta por duas funções contínuas em $\mathbb{R}\setminus\{0\}$

$$-\frac{1}{x^2} = y \longrightarrow e^y = \varphi(x)$$

$$\chi(x) = x \operatorname{sen}\frac{1}{x} - \cos\frac{1}{x}$$

é contínua em $\mathbb{R}\setminus\{0\}$, é composta por funções trigonométricas, polinomiais e racionais.

b) $\varphi(x) = e^{-1/x^2}$

$$\lim_{x\to 0} e^{-1/x^2} = e^{-1/+\infty} = e^0 = 1$$

É prolongável por continuidade no ponto 0

$$\varphi(x)\begin{cases} 1 & \text{se } x=0 \\ e^{-1/x^2} & \text{se } x\neq 0 \end{cases}$$ é contínua em $\mathbb{R}$

$$\chi(x) = x \operatorname{sen}\frac{1}{x} - \cos\frac{1}{x}$$

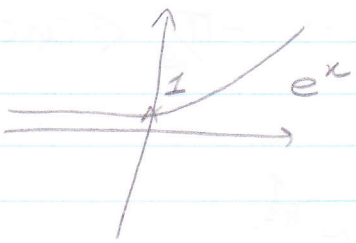
$$\lim_{x\to 0} x\cdot \operatorname{sen}\frac{1}{x} - \cos\frac{1}{x} = 0\cdot \operatorname{sen}(\pm\infty) - \cos(\pm\infty)$$

$= -\cos(\pm\infty)$ ← -1, 0 é descontínua

Pos

c) $\varphi(x)$ e $\chi(x)$ são limitadas

$\varphi(x) = e^{-1/x^2}$



$0 < e^{1/x} \leq e \iff$

$\iff 0 < e^{1/x \cdot 1/x} \leq e^{1/x} \iff$

$\iff 0 < e^{-1/x^2} \leq e^{1/x} \leq e$

Logo $\varphi(x)$ é limitada

$\chi(x) = x \operatorname{sen}\frac{1}{x} - \cos\frac{1}{x}$

limitada $[-1, 1]$

$-1 \leq \operatorname{sen}\frac{1}{x} \leq 1 \iff -x \leq x \operatorname{sen}\frac{1}{x} \leq x \iff$

$\iff -x - \cos\frac{1}{x} \leq x \operatorname{sen}\frac{1}{x} - \cos\frac{1}{x} \leq x - \cos\frac{1}{x}$

6.

a) $f(x) = \frac{\sqrt{x}}{x-1}$

$D = \{x : x-1 \neq 0 \wedge x \geq 0\}$

$= \{x : x \neq 1 \wedge x \geq 0\}$

$= [0, 1[\cup]1, +\infty[= \mathbb{R}_0^+ \setminus \{1\}$

b) $\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} \frac{\sqrt{x}}{x-1} = \frac{1}{+\infty} = 0$

$x_n = n$

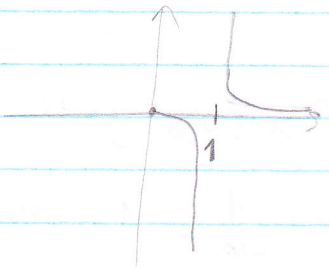
$\lim \frac{n}{n+1} = \frac{1}{1+\frac{1}{n}} = \frac{1}{1} = 1$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{\sqrt{x}}{\underbrace{x-1}_{0^-}} = \frac{1}{0^-} = -\infty$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \frac{\sqrt{x}}{\underbrace{x-1}_{0^+}} = \frac{1}{0^+} = +\infty$$

c)

$f(0) = 0$



$D' = \mathbb{R}$

→ porque quando $f(1^-) = -\infty$ e $f(1^+) = +\infty$

→ $\lim_{x\to+\infty} f(x) = 0$

mas $f(x)$ atinge 0 porque $f(0) = 0$

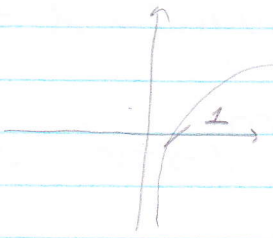
d)

7. $f(x) = \log\log(1+x)$ $\quad g(x) = \sqrt{x} \cdot \operatorname{sen}\frac{1}{x^2}$

$D =]0, +\infty[$

a) $\log(1+x) > 0 \Leftrightarrow 1+x > 1 \Leftrightarrow$

$\Leftrightarrow x > 0$



A função é contínua em todo o seu domínio, porque é composta por funções logarítmicas

$g(x)$ é composta por funções contínuas ($\sqrt{x}$) e $\operatorname{sen}\frac{1}{x^2}$ logo é contínua no seu domínio $]0,+\infty[$

b)

$$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} \log\log(1+x) =$$

$$= \log\log(+\infty) = \log(+\infty) = +\infty$$

$$\lim_{x\to+\infty} g(x) = \lim_{x\to+\infty} \sqrt{x}\cdot \operatorname{sen}\frac{1}{x^2} =$$

$$= 1\cdot \operatorname{sen}\frac{1}{+\infty} = 1\cdot \operatorname{sen}(0) = 1\cdot 0 = 0$$

c) $\lim_{x\to 0} \log\log(1+x) =$

$= \log\log(1) = \log 0 = -\infty$ não é prolongável por continuidade a 0

$$\lim_{x\to 0} \sqrt{x}\cdot \operatorname{sen}\frac{1}{x^2} = 0\cdot \operatorname{sen}\frac{1}{0}$$ (pNE)

d) Contradomínio $f(x) = \mathbb{R}$

quando $x\to 0$ $f(x)\to -\infty$

$\lim_{x\to 0} \log\log(x+1) = \log\log 1 = \log 0 = -\infty$

e quando $x\to+\infty$ $f(x)\to+\infty$

$\lim_{x\to+\infty} \log\log(x+1) = \log\log(+\infty) = \log(+\infty) = +\infty$

8. $f(x) = \begin{cases} -e^{1/x}, & x<0 \\ \log \frac{1}{1+x^2}, & \text{se } x>0 \end{cases}$

a)

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} -e^{1/x} =$$

$$= -e^{1/-\infty} = -e^0 = -1$$

$$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} \log \frac{1}{1+x^2} =$$

$$= \log \frac{1}{+\infty} = \log 0 = -\infty$$

b) sim: no ramo $x<0$ a função é exponencial, logo é contínua.

No ramo $x>0$ a função é logarítmica, e também contínua

c) $\lim_{x\to0^+} \log \frac{1}{1+x^2} = \log 1 = 0$

$$\lim_{x\to0^-} -e^{1/x} = -e^{-\infty} =$$

d)

9.

a) $g : [0, +\infty[\longrightarrow \mathbb{R}$ contínua

$$e(x) = g(1 - x^2)$$

$$1 - x^2 \geq 0 \iff -x^2 \geq -1 \iff$$

$$\iff x^2 \leq 1 \iff -1 \leq x \leq 1$$

Weierstrass ① $e(x) = [-1, 1] \rightarrow$ contínua

$$\min e(x) = e(-1) = e(1) = g(0)$$

$$\max e(x) = e(0) = g(1)$$

b) se $g(x)$ fosse $g :]0, +\infty[\longrightarrow \mathbb{R}$ contínua

$$1 - x^2 > 0 \iff -1 < x < 1$$

Logo não iria existir $e(-1) = e(1)$ que seria o máximo da função

Como $g(0)$ não fazia parte do domínio de $e(x)$, logo não ia existir mínimo de $e(x)$

10. $g :]a, b[\longrightarrow \mathbb{R}$ contínua

a b

$$\lim_{x \to a} g(x) = -\lim_{x \to b} g(x) = -\infty$$

Como $-\infty < g < +\infty$, então tem pelo menos um zero ($\exists c \in]a, b[$ tq $g(c) = 0$)

então $h(c) = \operatorname{arctg}(g(c)^2) =$

$= \operatorname{arctg} 0 = 0$

Como $h(0) = \pi/2$ e h é contínua em $[0, c]$, logo h atinge todos os valores em $[0, \pi/2]$

portanto $h(x): [0, \pi/2] \to \mathbb{R}$

11)

$\sin^3 x + \cos^3 x = 0$ tem pelo menos uma raiz em $]0, \pi[$

→ a função é contínua no seu domínio

12. f: contínua em $\mathbb{R}$ $\lim_{x \to +\infty} f(x)$ } são finitos
$\lim_{x \to -\infty} f(x)$ }

a) Se $\lim_{x \to +\infty} f(x) = a$

$\lim_{x \to -\infty} f(x) = b$ $\quad b < f(x) < a$

domínio da função é $]b, a[$

b)

$a \cdot b < 0$ → Tem pelo menos um zero em $]a, b[$

$g(x) = \dfrac{1}{1 + [f(x)]^2}$ $\quad f(c) = 0$

máximo $g(x) = g(f(c)) = \dfrac{1}{1} = 1$

$= 0$

Ficha 8

1.

a) $\log(x \operatorname{sh} x)$ → > 0

b) $\arcsen(\operatorname{arctg} x)$

$\forall x \in]-1,1[$ ← → $\mathbb{R} \setminus \{-1,1\}$

É diferenciável em $\operatorname{arctg} \in]-1,1[$ e $x \in \mathbb{R} \setminus \{-1,1\}$

$\operatorname{arctg} x = -1 \Leftrightarrow x = \operatorname{tg}(-1)$ → $x \in]\operatorname{tg}(-1), \operatorname{tg}(1)[$

$\operatorname{arctg} x = 1 \Leftrightarrow x = \operatorname{tg}(1)$

$\operatorname{sen} 3\alpha = x$

$\arcsen x = 3\alpha$

$$\arcsen' x = \frac{1}{\sqrt{1+x^2}}$$

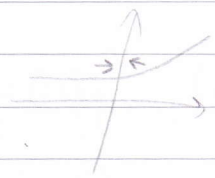
$$(\arcsen(\operatorname{arctg} x))' = \frac{1}{\sqrt{1+(\operatorname{arctg} x)^2}}$$

c) $\dfrac{e^x}{1+x}$ é diferenciável em $\mathbb{R} \setminus \{-1\}$

$$D' = \frac{e^x \cdot (1+x) - e^x \cdot 1}{(1+x)^2} = \frac{e^x + xe^x - e^x}{(x+1)^2}$$

2.

$$f(x) = \begin{cases} \dfrac{x}{1+e^{1/x}} & \text{se } x \neq 0 \\ 0 & \text{se } x = 0 \end{cases}$$



$$\lim_{x \to 0^-} \frac{\frac{x}{1+e^{1/x}} - 0}{x - 0} = \lim_{x \to 0^-} \frac{1}{1+e^{1/x}} = \frac{1}{1+e^{-\infty}} = \frac{1}{1+0} = 1$$

$$\lim_{x \to 0^+} \frac{1}{1+e^{1/x}} = \frac{1}{1+e^{+\infty}} = \frac{1}{1+\infty} = 0$$

3. se $x \neq 0$ $\quad f(x) = x\,\dfrac{1+e^{1/x}}{2+e^{1/x}}$

$$\lim_{x\to 0^-} \frac{x\,\frac{1+e^{1/x}}{2+e^{1/x}} - 0}{x - 0} = \lim_{x\to 0^-} \frac{1+e^{1/x}}{2+e^{1/x}} =$$

$$= \frac{1+e^{1/0^-}}{2+e^{1/0^-}} = \frac{1+0}{2+0} = \frac{1}{2}$$

$$\lim_{x\to 0^+} \frac{1+e^{1/x}}{2+e^{1/x}} = \lim_{x\to 0^+} \frac{\frac{1}{e^{1/x}}+1}{\frac{2}{e^{1/x}}+1} = \frac{0+1}{0+1} = 1$$

4.

$$f(x) = \begin{cases} x^2 \operatorname{sen}\left(\frac{1}{x}\right) & \text{se } x \neq 0 \\ 0 & \text{se } x = 0 \end{cases}$$

a)

É contínua em $\mathbb{R}\setminus\{0\}$

$$f'(x) = 2x \operatorname{sen}\frac{1}{x} - x^2 \cos\frac{1}{x} =$$

$$= x\left(2\operatorname{sen}\frac{1}{x} - x\cos\frac{1}{x}\right) \quad \forall x \in \mathbb{R}\setminus\{0\}$$

$$\lim_{x\to 0} f'(x) = \lim_{x\to 0} x\Big(\underbrace{2\operatorname{sen}\frac{1}{x}}_{\text{não existem}} - \underbrace{x\cos\frac{1}{x}}_{\text{não existem}}\Big)$$

b) $f(0) = 0$

$f(0^-) = x\left(2\operatorname{sen}\frac{1}{x} - x\cos\frac{1}{x}\right) = f(0^+)$

↳ tem derivada em 0

$f'(0) = 0$

5.

f diferenciável em $\mathbb{R}$

$f(0) = f(\pi) = 0$

$g = f(\operatorname{sen} x) + \operatorname{sen} f(x)$

$g(0) = f(\operatorname{sen} 0) + \operatorname{sen} f(0)$
$= f(0) + \operatorname{sen} 0 = 0 = f(0) = f(\pi)$

$g(\pi) = f(\operatorname{sen} \pi) + \operatorname{sen} f(\pi) =$
$= f(0) + \operatorname{sen} 0 = f(\pi) = f(0)$
$= 0$

Então $g'(0) + g'(\pi) = 0 + 0 = f'(0) + f'(\pi)$

6. $f: \mathbb{R} \to \mathbb{R}$, diferenciável

$(\operatorname{arctg} f(x) + f(\operatorname{arctg} x))' = (\operatorname{arctg} f(x))' + (f(\operatorname{arctg} x))'$

$= \frac{1}{1 + (f(x))^2} + (f(\operatorname{arctg} x))' \quad \forall x \in \mathbb{R}$

7. $g: \mathbb{R} \to \mathbb{R}$ duas vezes diferenciável

$e:]0, +\infty[\to \mathbb{R} \quad e(x) = e^{g(\log x)}$

g, g' e g''

$e'(1) \qquad e''(e)$

$e'(1) = e^{g(\log 1)} \cdot g(\log 1)' = g(0) \cdot e^{g(0)}$
$e''(e) = e^{g(\log e)} \cdot g(\log e)'' = g'(e)'' \cdot e^{g(1)}$

8. $f: \mathbb{R} \to \mathbb{R}$ $\quad f(x) = x^4 e^{-x}$
$g: \mathbb{R} \to \mathbb{R}$ diferenciável

$$(g \circ f)'(x) = g'(f(x)) \cdot g'(x) = g'(x^4 e^{-x}) \cdot g'(x)$$

9. $g: \mathbb{R} \to \mathbb{R}$ Estritamente monótona $\quad f: [-1, 1] \to \mathbb{R}$
$g(0) = 2$ Diferenciável
$g'(0) = \frac{1}{2}$ $\quad f(x) = g(\arcsen x)$

a) $f'(x) = g'(f(x)) \cdot g'(\arcsen x)$

$= g'(\arcsen x) \cdot \arcsen x = g'(\arcsen x) \cdot \frac{1}{\sqrt{1+x^2}}$
$\hookrightarrow \forall x \in [-1, 1]$

Como g é diferenciável, então f é diferenciável no domínio de arcsen x

$f'(0) = g'(\arcsen 0) \cdot g'(0)$

$= g'(0) \cdot g'(0) = \frac{1}{2} \frac{1}{2} = \frac{1}{4}$ $\quad \arcsen 0 = x \Leftrightarrow \sen x = 0 \Leftrightarrow x = k\pi = 0$

b) $f(x) = g(\arcsen x) \quad \forall x \in [-1, 1]$

a função arcsen x é injectiva no intervalo $[-1, 1]$

Como g(x) é estritamente monótona, g(arcsen x) é injectiva

$(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(0)} = \frac{1}{\frac{1}{2}} = 2$

como g é injectiva

$f(x) = 2 \quad f(0) = 2$
$g(\arcsen x) = 2$
$g(0) = 2$

10. $f: \mathbb{R} \longrightarrow]-1,1[$ diferenciável injectiva

$f(2) = 0$

$f'(2) = 2$

$g(x) = \arccos(f(x))$

a) $\arccos(x)$ é monótona e injectiva e $f(x)$ é injectiva.

logo $g(x)$ é injectiva

$\arccos(x)$ é diferenciável para $\forall x \in]-1,1[$
$f(x)$ é diferenciável e toma valores no intervalo $[-1,1]$

logo $g(x)$ é diferenciável

$$g'(2) = (\arccos(f(2)))' = \frac{1}{\sqrt{1-(f(2))^2}} = \frac{1}{\sqrt{1}} = 1$$

$$(g^{-1})'(\pi/2) = \frac{1}{g'(g^{-1}(\pi/2))} = \frac{1}{2}$$

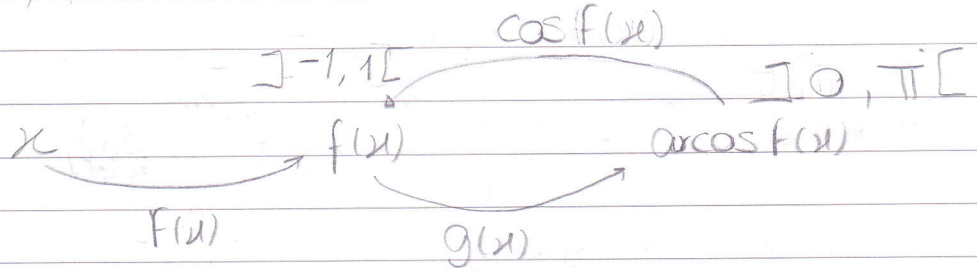
$g^{-1}(\pi/2) \Leftrightarrow g(x) = \frac{\pi}{2} \Leftrightarrow f(x) = 0 \Leftrightarrow x = 2$

$g(x) = \frac{\pi}{2} \Leftrightarrow \arccos f(x) = \frac{\pi}{2} \Leftrightarrow \cos\frac{\pi}{2} = 0$

$f(x) = 0 \Leftrightarrow x = 2$

b)

(g^{-1})



$(g^{-1}):]0,\pi[\longrightarrow]-1,1[$

Não tem máximo nem mínimo, mas é limitada

11. $sh(x) = \frac{e^x - e^{-x}}{2}$ $\quad ch(x) = \frac{e^x + e^{-x}}{2}$

a)

i) $ch^2 x - sh^2 x = 1$

$\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 =$

$\frac{(e^x - e^{-x})^2 - (e^x + e^{-x})^2}{4} = \frac{2e^0 - (-2e^0)}{2} =$

$= \frac{2+2}{4} = 1$ cqd

ii) $sh(x+y) = shx\,chy + chx\,shy$

$sh(x+y) = \frac{e^{x+y} - e^{-(x+y)}}{2}$

$\frac{e^x - e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2}$

$\frac{(e^x - e^{-x})(e^y + e^{-y}) + (e^x + e^{-x})(e^y - e^{-y})}{4} = \frac{e^{x+y} + e^{x-y} - e^{-x+y} - e^{-y-x} + e^{x+y} - e^{x-y} + e^{y-x} - e^{-x-y}}{4}$

$\frac{e^{x+y} + e^{-(x+y)}}{2}$

iii) $ch(x+y) = chx\,chy + shx\,shy$

$\frac{e^x + e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x - e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2} =$

$= \frac{2e^{x+y} + 2e^{-(x+y)}}{2} = \frac{e^{x+y} + e^{-(x+y)}}{2}$ cqd

$\frac{e^{2x} - e^{-2x}}{2}$ iv) $sh(2x) = 2shx\,chx$

$2\left(\frac{e^x - e^{-x}}{2} \cdot \frac{e^x + e^{-x}}{2}\right) = \frac{(e^x - e^{-x})(e^x + e^{-x})}{2} =$

$= \frac{e^{2x} - e^{-2x}}{2}$ cqd

$\frac{e^{2x} + e^{-2x}}{2}$ v) $ch(2x) = ch^2x + sh^2x$

$\left(\frac{e^x + e^{-x}}{2}\right)\left(\frac{e^x + e^{-x}}{2}\right) + \left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^x - e^{-x}}{2}\right) =$

$= \frac{e^{2x} + e^0 + e^0 + e^{-2x} + e^{2x} - e^0 - e^0 + e^{-2x}}{4} =$

$= \frac{2e^{2x} + 2e^{-2x}}{2} = \frac{e^{2x} + e^{-2x}}{2}$ cqd

b)

sh é impar	ch é par
$sh(x) = -sh(-x)$	$ch(x) = ch(-x)$
$-sh(-x) = -\left(\frac{e^{-x} - e^x}{2}\right) =$	$Ch(-x) = \frac{e^{-x} + e^x}{2} = \frac{e^x + e^{-x}}{2}$
$= \frac{-e^{-x} + e^x}{2} = \frac{e^x - e^{-x}}{2}$	cqd
cqd	

d) São ambos contínuas em $\mathbb{R}$

$$D\,sh(x) = \frac{(e^x - e^{-x})' \cdot 2 - (e^x - e^{-x}) \cdot 2'}{4} =$$

$$= \frac{2(e^x + e^{-x}) - 0}{4} = \frac{e^x + e^{-x}}{2} = ch(x)$$

$$D\,ch(x) = \frac{(e^x + e^{-x})' 2 - (e^x + e^{-x}) \cdot 2'}{4} =$$

$$= \frac{2(e^x - e^{-x})}{4} = \frac{e^x - e^{-x}}{2} = sh(x)$$

e) $sh(x)$

É negativa quando
$e^{-x} > e^x \iff$
$\iff -x > x \iff$
$\iff -x - x > 0 \iff$
$\iff -2x > 0 \iff x < 0$

É positiva quando
$e^{-x} < e^x \iff$
$\iff x > 0$

$sh(0) = 0$

zero ímpar

Como $D\,sh(x) = ch(x)$
então sh é estritamente crescente

Derivada decrescente

Derivada crescente

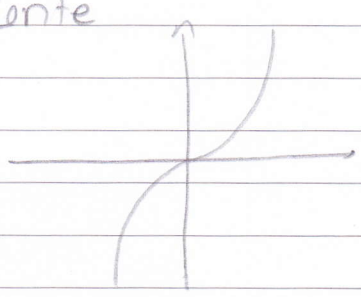
$ch(x)$

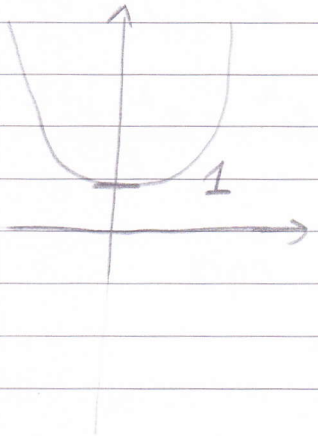
nunca é negativa,
não tem zeros

$D\,ch(x) = 0 \iff$
$sh(x) = 0 \iff$
$x = 0$

ch é estritamente decrescente em $]-\infty, 0]$
ch é estritamente crescente em $]0, +\infty[$
tem extremo (mínimo) quando $x = 0$
Par

$$\min = \frac{e^0 + e^{-0}}{2} = 1$$





f) $(\operatorname{sh} x^{-1}) = \operatorname{argsh}(x)$ | $(\operatorname{ch} x^{-1}) = \operatorname{argch}$

$x = \operatorname{sh} y$ sse $y = \operatorname{argsh}(x)$ | $x = \operatorname{ch} y$ sse $y = \operatorname{argch}$

não percebi!

12. $f :]0, 1[\to \mathbb{R}$ diferenciável

$f\left(\frac{1}{n+1}\right) = 0 \quad \forall n \in \mathbb{N}_1$

a) $\forall n \geq 2$ f tem max em $\left[\frac{1}{n+1}, \frac{1}{n}\right]$

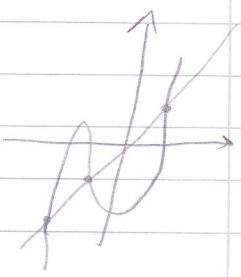
$n=2$ $f\left(\frac{1}{3}\right) = 0$ tem max em $\left[\frac{1}{3}, \frac{1}{2}\right]$?

b) f é necessariamente limitada

c) f' tem necessariamente infinitos zeros

13. $f : \mathbb{R} \to \mathbb{R}$ duas vezes diferenciável
corta a recta $x = y$ em 3 pontos

então f'' tem pelo menos um zero



$\exists \quad a < b < c$ tal que $f(a) = a$, $f(b) = b$, $f(c) = c$ → 2 vezes dif.

→ $\exists\, z \in \mathbb{R} : f''(z) = 0$

ou $g(x) = f(x) - x$

$g(a) = g(b) = g(c)$

Aplicando o Teorema de Lagrange:

pelo T. de Rolle $\exists c_1 \in]a,b[$ tq $g'(c_1) = 0$

→ a $[a, b]$ $\exists\, d \in]a, b[$

$$tq\ f'(d) = \frac{f(b) - f(a)}{b - a} = 1$$

$\exists c_2 \in]a,b[$ tq $g'(c_2) = 0$

→ a $[b, c]$ $\exists\, e \in]b, c[$

$$tq\ f'(e) = \frac{f(c) - f(b)}{c - b} = 1$$

$\exists c_3 \in]c_1, c_2[$ tq $g''(c_3) = 0$

Aplicando o teorema de Rolle a f' em $[d, e]$

$\exists\, z \in [d, e]$ tq $f'(z) = 0$

$g'(x) = f'(x) - 1$

$g''(x) = f''(x)$

c.q.d.

14. $3x^2 - e^x$ tem exactamente 3 zeros

$(3x^2 - e^x)' = 2x - e^x$

ficha 9

1. $f: \mathbb{R}_0^+ \to \mathbb{R}$ diferenciável e $f(n) = (-1)^n \quad \forall n \in \mathbb{N}$

$-1 \quad 1$

→ a sua derivada não tem limite no infinito

$\lim f'(n)$ Não existe

↳ $\lim ((-1)^n)'$

2. $f(0) = 0$ derivada é uma função crescente

$g(x) = \frac{f(x)}{x}$ é crescente em $\mathbb{R}^+$

Aplicando o T de Lagrange a $f(x)$ no intervalo $]0, x[\quad \forall x \in \mathbb{R}_+$

$\exists c \in]0, x[$ tq $\frac{f(x) - f(0)}{x - 0} = f(c)$

$\iff \frac{f(x)}{x} = f(c)$

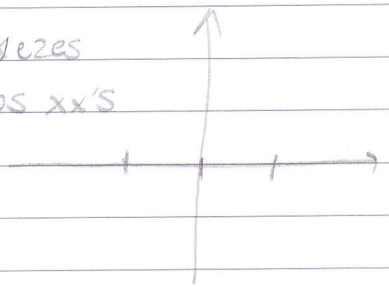
Como $c > 0$ então em $\mathbb{R}_+$ $F(c) = g(x)$

3. Se $f \in C^1(\mathbb{R})$ e $f(x) = x^2$ tem 3 soluções
(uma negativa, uma positiva, uma nula)

↓

então $f'(x)$ tem pelo menos um zero

- Se é derivável uma vez
- Passa 3 vezes pelo ref. dos xx's



pelo ~~teorema de Rolle~~ teorema de Rolle

para qualquer intervalo $[a,b]$ tq $f(a) = f(b) = 0$ $a < 0$ $b > 0$

$f(a) = f(b) = f(0) = 0$

então $\exists d \in]a, d[: f'(d) = 0$

4.

a) $D_{\frac{x}{x^2+1}} = \frac{1.(x^2+1) - x(2x+0)}{(x^2+1)^2} = \frac{x^2+1-2x^2}{(x^2+1)^2} =$

$= \frac{-(x^2-1)}{(x^2-1)^2} = \frac{-1}{x^2-1}$ $x \neq 1$ e -1

→ Não tem extremos

→ A função é | decrescente em $]-\infty, -1[\cup]1, +\infty[$
crescente em $]-1, 1[$

b) $\frac{1}{x} + \frac{1}{x^2} = \frac{1.1 - 0.x}{x^2} + \frac{1.2x - 0.x^2}{x^4} =$

$= \frac{1}{x^2} + \frac{2x}{x^4} = \frac{x^2+2x}{x^4} = \frac{x+2}{x^3}$ $x \neq 0$

→ tem zero em $x = -2$

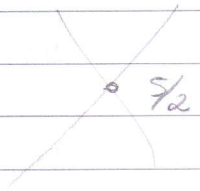
	$-\infty$		-2		0	$+\infty$
$x+2$	$-$		0	$+$	$+$	$+$
x^3	$-$		$-$	$-$	0	$+$
	$+$		0	$-$	ND	$+$

→ A função é
crescente de $]-\infty, -2[$
tem um máximo local em -2
é decrescente em $]-2, 0[$
e crescente em $]0, +\infty[$

c) $|x^2-5x+6|' =$

$(x^2-5x+6)' = 2x-5 \qquad x = \frac{5}{2}$

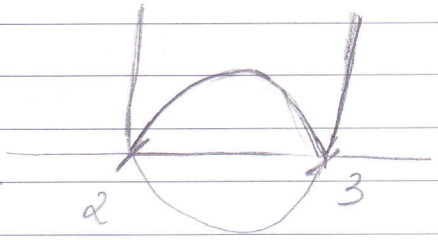
$(-x^2+5x-6)' = -2x+5 \qquad x = \frac{-5}{-2}$



$x^2-5x+6=0 \iff x = \frac{5 \pm \sqrt{25-4.1.6}}{2} \iff x = \frac{5 \pm \sqrt{1}}{2} \iff$

$\iff x = \frac{6}{2} \vee x = \frac{4}{2} \iff$

$\iff x = 3 \vee x = 2$

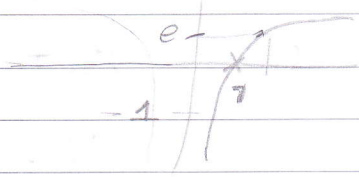


		5/2	
$2x-5$	$-$	0	$+$
$-2x+5$	$+$	0	$-$
	$-$	0	$-$

d) $(x \log x)' = x \cdot \frac{1}{x} + \log x = 1 + \log x$

$\forall x \in \mathbb{R}^+$

$1+\log x = 0 \iff -\log x = 1$



$\iff \log \frac{1}{x} = 1 \iff \frac{1}{x} = e \iff$

$\iff 1 = ex \iff \frac{1}{e} = x$

a função é | decrescente em $]0, \frac{1}{e}[$ | tem mínimo em $x = \frac{1}{e}$
| crescente em $]\frac{1}{e}, +\infty[$ |

e) $\left(\overbrace{e^{-x^2}}^{\text{sempre}>0}\right)' = -2x \cdot e^{-x^2}$

sempre < 1

		0	
$-2x$	+	0	−
e^{-x^2}	+	+	+
	+	0	−

$-2x \cdot e^{-x^2} = 0 \Leftrightarrow x = 0$

a função | tem máximo em $x = 0$
é crescente em $]0, +\infty[$
é decrescente em $]-\infty, 0[$

f) $\left(\frac{e^x}{x}\right)' = \frac{e^x - e^x \cdot x}{x^2}$ $\quad \forall x \neq 0$

$\frac{e^x - e^x \cdot x}{x^2} = 0 \Leftrightarrow e^x - e^x \cdot x = 0 \Leftrightarrow e^x = e^x \cdot x$

$\Leftrightarrow \frac{e^x}{e^x} = x \Leftrightarrow x = 1$

f é | decrescente em $]-\infty, 0[\cup]0, 1[$
crescente em $]1, +\infty[$

g) $(xe^{-x})' = x \cdot (-e^{-x}) + e^{-x} = -xe^{-x} + e^{-x}$

$\Leftrightarrow e^{-x}(-x + 1) = e^{-x}(1 - x)$

$e^{-x}(1 - x) = 0 \Leftrightarrow e^{-x} = 0 \vee 1 - x = 0 \Leftrightarrow$

PF

$\Leftrightarrow x = -1$

		−1	
e^{-x}	+	+	+
$1 - x$	+	0	−
	+	MÁX	−

f é | crescente em $]-\infty, -1[$
decrescente em $]-1, +\infty[$
tem máximo em $x = -1$

h) $(\arctg x - \log\sqrt{1+x^2})' = \frac{1}{\sqrt{1+x^2}} -$

$\left(\log \frac{1}{\sqrt{1+x^2}}\right)' =$

$\left(\frac{1}{\sqrt{1+x^2}}\right)' = \frac{0 \cdot \sqrt{1+x^2} - 1 \cdot \frac{1}{2\sqrt{1+x^2}}}{1+x^2} = -\frac{1}{2\sqrt{1+x^2}\cdot(1+x^2)}$

$(\sqrt{1+x^2})' = \frac{(1+x^2)^{-1/2}}{2} = \frac{1}{2\sqrt{1+x^2}}$

5. a) $f: \mathbb{R} \to \mathbb{R} \quad f(x) = |x| e^{-\frac{x^2}{2}}$

$\lim_{-\infty} |x| e^{-\frac{x^2}{2}} = +\infty \cdot e^{-\infty} = +\infty \cdot 0$ ($|x| \to +\infty$)

$\lim_{-\infty} \frac{|x|}{e^{x^2/2}}$ pela regra de cauchy $= \lim_{x \to -\infty} \frac{1}{\underbrace{x e^{x^2/2}}_{\to \infty}} = 0$

$\lim_{+\infty} |x| e^{-x^2/2} = +\infty \cdot e^{-\infty} = +\infty \cdot 0$ ($|x| \to +\infty$)

$\lim_{+\infty} \frac{|x|}{e^{x^2/2}}$ pela regra do cauchy $= \lim_{x \to +\infty} \frac{1}{\underbrace{x e^{x^2/2}}_{+\infty}} = 0$

b) em $\mathbb{R} \setminus \{0\}$

$(|x| e^{-\frac{x^2}{2}})' = \begin{cases} \frac{1 - x^2}{e^{x^2/2}} & \text{se } x > 0 \\ \frac{x^2 - 1}{e^{x^2/2}} & \text{se } x < 0 \end{cases}$

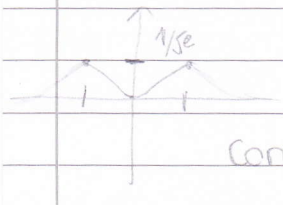
c) $\frac{1-x^2}{e^{x^2/2}}$

		-1		0		1	
$1-x^2$	$-$	0	$+$		$+$	0	$-$
$e^{x^2/2}$	$+$		$+$		$+$		$+$
φ	$-$	0	$+$		$+$	0	$-$
x^2-1	$+$	0	$-$		$-$	0	$+$
φ	$+$	0	$-$		$-$	0	$+$

crescente em $]-\infty,-1[$
tem um máximo em $x=-1$
decrescente em $]-1,0[$
crescente em $]0,1[$
tem um máximo em $x=1$
decrescente em $]1,+\infty[$

d) maximos $x=-1$ | $x=1$

$|-1|\, e^{\frac{-(-1)^2}{2}} = 1.e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$ | $|1|\, e^{-\frac{1^2}{2}} = 1.e^{-1/2} = \frac{1}{\sqrt{e}}$



Como $\lim_{x\to-\infty} |x|e^{-x^2/2} = \lim_{x\to+\infty} |x|e^{-x^2/2} = 0$

Contradominio de $g(x) =]0, 1/\sqrt{e}]$

6. $g(x) = \begin{cases} e^x + \alpha x + \beta & \text{se } x \le 0 \\ \operatorname{arctg}(e^x + e^{-x} - 1) & \text{se } x > 0 \end{cases}$

a)

$\lim_{x\to0^+} \operatorname{arctg}(e^x + e^{-x} - 1) = \lim_{x\to0^+} \operatorname{arctg}(e^{0^+} + e^{0^-} - 1) =$ ($e^{0^+} > 1$, $e^{0^-} < 1$)

$= \operatorname{arctg}(1) = \pi/4$

$e^0 + \alpha.0 + \beta = \pi/4 \Leftrightarrow 1 + \beta = \pi/4 \Leftrightarrow \beta = \pi/4 - 1 \Leftrightarrow \beta = \frac{\pi}{4} - \frac{4}{4}$

$\Leftrightarrow \beta = \frac{\pi - 4}{4}$

$\beta = 4 \qquad \alpha = -1$

b)

$$\lim_{x \to -\infty} g(x) = \lim_{x \to -\infty} \underset{\to 0}{e^x} - \underset{\to +\infty}{x} + 4 =$$

$$\lim_{x \to -\infty} e^x - e^{\log x} + 4$$

$$\lim_{x \to +\infty} g(x) = \lim_{x \to +\infty} \operatorname{arctg}(\underset{\to +\infty}{e^x} + \underset{\to 0}{e^{-x}} - 1)$$

c) $(e^x - x + 4)' = e^x - 1 \qquad x \leq 0$

$$(\operatorname{arctg}(e^x + e^{-x} - 1))' = \frac{1}{1 - (e^x + e^{-x} - 1)^2} \cdot (e^x + e^{-x} - 1)'$$

$$= \frac{1}{1 - (e^x + e^{-x} - 1)^2} \cdot (e^x - e^{-x})$$

d) $e^x - 1$

7. $f: \mathbb{R} \to \mathbb{R}$ $\quad |x|\, e^{-|x-1|}$

a) $\lim_{x \to -\infty} |x|\, e^{-|x-1|} = \frac{|x|}{e^{|x-1|}}$ pela regra de Cauchy

$\lim_{x \to -\infty} = \frac{1}{e^{|x-1|}}$ se $x \neq 0 \iff$

$\iff \lim_{x \to -\infty} \frac{1}{e^{+\infty}} = \frac{1}{+\infty} = 0 = \lim_{x \to +\infty} |x|\, e^{-|x-1|}$

$$|x-1|' = \begin{cases} 1 & \text{se } x > 0 \\ -1 & \text{se } x < 0 \end{cases}$$

b) em $x \neq 0$

$$(|x|\, e^{-|x-1|})' = \begin{cases} -1 \cdot e^{-|x-1|} + |x| \cdot e^{-|x-1|} & x < 0 \\ 1 \cdot e^{-|x-1|} + |x| \cdot (-e^{-|x-1|}) & x > 0 \end{cases} =$$

$$= \begin{cases} -e^{-|x-1|} + |x|\, e^{-|x-1|} \\ e^{-|x-1|} - |x|\, e^{-|x-1|} \end{cases} = \begin{cases} e^{-|x-1|}(-1-x) & x < 0 \\ e^{-|x-1|}(1-x), & x > 0 \end{cases}$$

c)

		-1		0		1	
$e^{-\lvert x-1\rvert}$	$+$	$+$	$+$	$+$	$+$	$+$	$+$
$(-x-1)$	$+$	0	$-$	$-$	$-$	$-$	$-$
$(-x+1)$	$+$	$+$	$+$	$+$	$+$	0	$-$
Q	$+$	0	$-$	$\nexists$	$-$	0	$+$
	↗	MAX	↘	MIN	↘	MAX	↗

$x = -1$

$|-1| e^{-|-1-1|} = 1 \cdot e^{-2} = e^{-2}$

$x = 1$

$|1| e^{-|1-1|} = e^0 = 1$ → máximo absoluto

$x = 0$

$0 \cdot e^1 = 0$ → mínimo absoluto

d) $[0, 1]$

3. $f(x) = x + 2\,\text{arctg}|x| \quad \mathbb{R} \to \mathbb{R}$

a) $\lim_{x \to -\infty} x + 2\,\text{arctg}|x| = -\infty + 2 \cdot \frac{\pi}{2} = -\infty + \pi = -\infty$

$\lim_{x \to +\infty} x + 2\,\text{arctg}|x| = +\infty + \frac{2\pi}{2} = +\infty + \pi = +\infty$

b) $\mathbb{R} \setminus \{0\}$

$(x + 2\,\text{arctg}\,x)' = 1 + 2 \cdot \frac{1}{1-x^2} = 1 + \frac{2}{1-x^2} \qquad x > 0$

$|x|' = 1$

$(x + 2\,\text{arctg}(-x))' = 1 + 2 \cdot \frac{1}{1-(-x)^2} \times (-x)' \qquad x < 0$

$= \frac{1-2}{1+x^2}$

$|x| = -1$

$x = 0$

$\lim_{x \to 0^+} f(x)' = 3 \neq \lim_{x \to 0^-} f(x)'$

$x<0$

c)

$$1-\frac{2}{1+x^2}=0 \Leftrightarrow \frac{2}{1+x^2}=1 \Leftrightarrow 2=1+x^2 \Leftrightarrow$$

$$\Leftrightarrow 2-1=x^2 \Leftrightarrow x^2=1 \Leftrightarrow x=1 \wedge x=-1$$

		-1	0	1	
$1-\frac{2}{1+x^2}$	+	0	−	0	+
$1+\frac{2}{1+x^2}$	+	+	+	+	+
$f'(x)$	+	0	-NE+	+	+

$$1+\frac{2}{1+x^2}=0 \Leftrightarrow \frac{2}{1+x^2}=-1 \Leftrightarrow 2=-1-x^2 \Leftrightarrow$$

$\Leftrightarrow 3=-x^2 \Leftrightarrow x^2=-3$ PF

f é crescente em $]-\infty,-1[$
tem máximo em $x=-1$ → máximo local
é decrescente em $]-1,0[$
tem um vértice em $x=0$ → mínimo local
é crescente em $]0,+\infty[$

$$f(0)=0+2\arctg 0=2.0=0$$

$$f(-1)=-1+2\arctg(-1)=\frac{\pi}{2}-1$$

d) $f:]-\infty,0] \longrightarrow]-\infty, \pi/2-1$

9. $g(0) = g'(0) = 0$ g' é estritamente monótona

$e(x) = 2\,tg(g(x)) - g(x)$

então $e(0)$ é extremo local de e

$e(0)' = 0$

$e'(x) = (2tg(g(x)) - g(x))' = (2tg(g(x)))' - g(x)' =$

$= 2 \cdot (tg(g(x)))' - g(x)' = 2 \cdot tg'(g(x)) \cdot g(x)' - g(x)'$

no ponto $x = 0$

$2 \cdot tg'(0) \cdot 0 - 0 = 0$

Como $x=0$ é zero da derivada, 0 é extremo local de e

10. $f: V_\varepsilon(0)$ dif em $V_\varepsilon(0) \setminus \{0\}$

e $xf'(x) > 0 \quad \forall x \in V_\varepsilon(0) \setminus \{0\}$

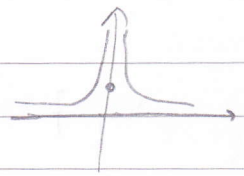
a) f contínua em $x = 0$

Prove que $f(0)$ é um extremo

$x > 0 \ \vee \ f'(x) > 0$	$x < 0 \ \vee \ f'(x) < 0$
a função é crescente à esquerda do 0	a função é decrescente à direita do zero

Logo 0 é um máximo local da função

Se existir $f'(0) = 0$

b) por exemplo 

não é contínua em $x=0$, não se pode concluir que este é extremo

11.

a) $\lim_{x\to 0} \frac{a^x - b^x}{x}$ pela regra de Cauchy

$$\lim_{x\to 0} \frac{x a^{x-1} - x b^{x-1}}{1} = \lim_{x\to 0} x(a^{x-1} - b^{x-1})$$

$$= 0(a^{-1} - b^{-1}) = 0$$

b) $\lim_{x\to +\infty} \frac{\log(x + e^x)}{x}$ pela regra de Cauchy

$$\lim_{x\to +\infty} \frac{(x+e^x)'}{(x+e^x)} = \lim_{x\to +\infty} \frac{1+e^x}{x+e^x}$$ pela regra de Cauchy

$$= \lim_{x\to +\infty} \frac{e^x}{1+e^x} = \lim_{x\to +\infty} \frac{1}{\frac{1}{e^x}+1} = \frac{1}{0+1} = 1$$

c) $\lim_{x\to 1} \log x \cdot \log\log x = \lim_{x\to 1} \frac{\log\log x}{\frac{1}{\log x}} =$

pela regra de Cauchy $= \lim_{x\to 1} \frac{\frac{1}{\log x}\cdot\frac{1}{x}}{\frac{-1}{(\log x)^2}\cdot\frac{1}{x}} =$

$$= \lim_{x\to 1} -\frac{\log^2 x}{\log x} = \lim_{x\to 1} -\log x = 0$$

d) $\lim_{x \to 0^+} \frac{e^{-1/x}}{x}$ ($e^{-1/x} \to -\infty$) pela regra de Cauchy =

$= \lim_{x \to 0^+} \frac{1/x^2 \cdot e^{-1/x}}{1} =$

$=$

e) $\lim_{x \to 0^-} \frac{e^{-1/x}}{x}$

f) $\lim_{x \to 1^+} x^{\log \log x} = \lim_{x \to 1^+} (e^{\log x})^{\log \log x} =$

$= \lim_{x \to 1^+} e^{\log x \log \log x} = e^0 = 1$

g) $\lim_{x \to +\infty} x^{\frac{1}{x-1}} =$

12.

a) $\lim_{x \to 0} \frac{10^x - 5^x}{x}$ pela r de Cauchy

$= \lim_{x \to 0} \frac{x 10^{x-1} - x 5^{x-1}}{1} = \frac{0.10^{-1} - 0.5^{-1}}{1} = 0$

b) $\lim_{x \to 0^+} \frac{x^2 \operatorname{sen} \frac{1}{x}}{\operatorname{sen} x}$ pela regra de cauchy

$$\lim_{x \to 0^+} \frac{\left(x^2 \operatorname{sen} \frac{1}{x}\right)'}{-\cos x} = \lim_{x \to 0^+} \frac{2x \operatorname{sen} \frac{1}{x} + \cos \frac{1}{x}}{-\cos x}$$

C.A

$$\left(x^2 \operatorname{sen} \frac{1}{x}\right)' = 2x \cdot \operatorname{sen} \frac{1}{x} + x^2 \cdot \frac{1}{x^2} \cos \frac{1}{x}$$

$$= 2x \cdot \operatorname{sen} \frac{1}{x} + \cos \frac{1}{x}$$

13.

a) $\lim_{x \to +\infty} \frac{2^x}{x^2} = +\infty$

b) $\lim_{x \to -\infty} \frac{2^x}{x^2} = 0$

14.

a) $\lim_{x \to 0^+} (\operatorname{sen} x)^{\operatorname{sen} x} = \lim_{x \to 0^+} e^{\log \operatorname{sen} x^{\operatorname{sen} x}}$

$$= \lim_{x \to 0^+} e^{\operatorname{sen} x \cdot \log \operatorname{sen} x} = e^0 = 1$$

$\lim_{x \to 0^+} \operatorname{Sen} x \cdot \log(\operatorname{sen} x) = \lim_{x \to 0^+} \frac{\log \operatorname{sen} x}{\frac{1}{\operatorname{sen} x}} = \frac{\frac{-\cos x}{\operatorname{sen} x}}{\frac{-\cos x}{\operatorname{sen} x^2}} =$

$= \frac{\operatorname{sen} x^2 \cdot (-\cos x)}{\operatorname{sen} x \cdot (-\cos x)} = \lim_{x \to 0^+} \operatorname{sen} x = 0$

$(\log \operatorname{sen} x)' = \frac{-\cos x}{\operatorname{sen} x} \quad \left(\frac{1}{\operatorname{sen} x}\right)' = \frac{-\cos x}{\operatorname{sen} x^2}$

$+\infty^0$

b) $\lim_{x\to+\infty} (\log x)^{1/x} \Leftrightarrow \lim_{x\to+\infty} e^{\frac{1}{x}\log(\log x)}$ ($0.\infty$) $= e^0 = \underline{1}$

$\Leftrightarrow \lim_{x\to+\infty} \frac{1}{x}\log(\log x) = \lim_{x\to+\infty} \frac{\log(\log x)}{x} =$

cauchy $\frac{\infty}{\infty}$ ou $\frac{0}{0}$

$= \lim_{x\to+\infty} \frac{\frac{1}{\log x}\cdot\frac{1}{x}}{1} = \lim_{x\to+\infty} \frac{1}{x\log x} = \frac{1}{+\infty} = 0$

$+\infty.0$

15.

a) $\lim_{x\to+\infty} x\operatorname{sen}\frac{1}{x} = \lim_{x\to+\infty} \frac{\operatorname{sen}\frac{1}{x}}{\frac{1}{x}} =$

R. Cauchy $\frac{\frac{1}{x^2}\cdot(-\cos\frac{1}{x})}{\frac{1}{x^2}} = \lim_{x\to+\infty} -\cos\frac{1}{x} = -1$

b) $\lim_{x\to\frac{\pi}{4}} (\operatorname{tg} x)^{\operatorname{tg} 2x} = \lim_{x\to\frac{\pi}{4}} e^{\log \operatorname{tg} x \operatorname{tg} 2x} =$

$= \lim_{x\to\frac{\pi}{4}} e^{\operatorname{tg} 2x \cdot \log \operatorname{tg} x}$

$\lim_{x\to\pi/4} \operatorname{tg} 2x \cdot \log \operatorname{tg} x = \lim_{x\to\pi/4} \frac{\log \operatorname{tg} x}{\frac{1}{\operatorname{tg} 2x}}$ R. Cauchy $\frac{(\log \operatorname{tg} x)'}{\frac{1}{\cos^2 x \operatorname{tg} x}}$

$\lim_{x\to\pi/4}$

16) $\lim \left(\frac{1}{n}\right)^{\sin\frac{1}{n}}$

$$\lim_{x\to 0} x^{\sin x} \overset{0^0}{=} \lim_{x\to 0} e^{\log x^{\sin x}} = \lim_{x\to 0} e^{\sin x \log x}$$

C.A:

$$\lim_{x\to 0} \sin x \log x = \lim_{x\to 0} \frac{\log x}{\frac{1}{\sin x}} \quad \text{R. Cauchy}$$

$$\lim_{x\to 0} \frac{1/x}{\frac{-\cos x}{\sin x^2}} = \lim_{x\to 0} -\frac{\sin x^2}{x.\cos x} \quad \text{R. Cauchy}$$

$$\lim_{x\to 0} -\frac{1-\cos^2 x}{x\cos x} = \lim_{x\to 0} \frac{-1}{x\cos x} - \frac{\cos x}{x}$$

$$= -\frac{1}{0^+} - \frac{1}{1}$$

17.

a) Ordem 2, ponto 1

$f(x) = e^{2x}$ $f'(x) = 2e^{2x}$ $f''(x) = 2.2.e^{2x} = 4e^{2x}$

Mc Laurin: $f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!}$

$$= e^0 + 2e^0 x + \frac{4e^0 x^2}{2} = 1 + 2x + \frac{4x^2}{2} = 2x^2 + 2x + 1$$

Taylor: $f(x) = e^1 + 2e^1 x + \frac{4e^1 x^2}{2} = 2ex^2 + 2ex + e = e(2x^2 + 2x + 1)$

$f(x)=\log(1+x)$ $\quad f'(x)=\frac{1}{1+x}$ $\quad f''(x)=\frac{-1}{(1+x)^2}$

Mclaurin $\quad f(x)=f(0)+f'(0)x+\frac{f''(0)x^2}{2!}$

$=\log 1+\frac{1}{1}x+\frac{1}{2}x^2$

$=x+\frac{1}{2}x^2$

taylor $\quad F(x)=\log 2+\frac{1}{1+1}x+\frac{1}{4}x^2=\log 2+\frac{1}{2}x+\frac{1}{8}x^2$

$f(x)=\cos(\pi x)$ $\quad f'(x)=-\pi\operatorname{sen}(\pi x)$ $\quad f''(x)=-2\pi\cos(\pi x)$

Mclaurin $\quad f(x)=\cos 0-\pi\operatorname{sen}0\,x-\frac{2\pi\cos 0x^2}{2}=1-\pi x^2$

taylor $\quad F(x)=\cos\pi+(-\pi\operatorname{sen}\pi)+\frac{(-2\pi\cos\pi)}{2}x^2=-1+\pi x^2$

b)

18.

19. $\forall x\in\mathbb{R}\ \exists\xi_x$ entre 0 e x $\quad$ como $x\in[0,1]$ $e^{\xi_x}\frac{x^3}{6}<\underbrace{e^{\xi_x}}_{\leq 1}\frac{1}{6}<\frac{1}{6}$

$f(x)=f(0)+f'(0)x+\frac{f''(0)x^2}{2!}+f'''(\xi_x)\frac{x^3}{3!}$

$f(x)=e^{-x}$ $\quad \forall x\in\mathbb{R}$ $\quad e^{-x}=1-x+\frac{x^2}{2}-e^{-\xi_x}\cdot\frac{x^3}{3!}$

$f(x)'=-e^{-x}$ $\quad$ como $x\in[0,1]$ $\quad 0<\xi_x<x<1$

$f''(x)=+e^{-x}$ $\quad \left|e^{-x}-\left(1-x+\frac{x^2}{2}\right)\right|=\left|-e^{\xi_x}\cdot\frac{x^3}{3!}\right|=e^{\xi_x}\frac{x^3}{6}$ (>0, $\xi_x\in[0,1]$)

$f'''(x)=-e^{-x}$

TU AMAS-ME?!

20.

$$P_2(x) = g(0) + g'(0)x + \frac{g''(0)}{2!}x^2$$

$$g(0) = f(e^0) = f(1) = 2$$

$$f(1) + f'(1)(x-1) + \frac{f''(1)(x-1)^2}{2!} = 3 - x + 2(x-1)^2$$

(2 ↙, 1 ↖, $2 \cdot 2! = 4$ ↙)

$$= 2 - (x-1) + 2(x-1)^2$$

$$g'(x) = e^x f'(e^x) \Rightarrow g'(0) = e^0 f'(e^0) = f'(1) = -1$$

$$g''(x) = e^x f''(e^x) \cdot e^x + e^x \cdot f'(e^x)$$

$$\Downarrow$$

$$g''(0) = e^0 f''(e^0) \cdot e^0 + e^0 \cdot f'(e^0)$$

$$= 1 \cdot f''(1)\, 1 + 1 \cdot f'(1)$$

$$= (-1) + 4 = 3$$

$$P_2(x) = 2 + (-1) \cdot x + \frac{3}{2!}x^2$$

$$= 2 - x + \frac{3}{2}x^2$$

21. Se $f: \mathbb{R} \to \mathbb{R}$ e $f^n(x) = 0 \ \forall x \in \mathbb{R}$ então f tem grau inferior a n

$$P_n(x) = f(0) + \ldots + \frac{f^n(0)x^n}{n!}$$

($f^n(0) = 0$)

22. $I \in \mathbb{R}$ $]a;b[$

$f \in C^2(I)$

$\forall a \in I$

$$P_2(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + R_3(x) \quad \underset{h \to 0}{\Leftrightarrow} \quad f''(a) = \lim \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}$$

$$\Leftrightarrow \frac{f''(a)(x-a)^2}{2} = -f(a) - f'(a)(x-a) + f(x)$$

$$\Leftrightarrow f''(a)(x-a)^2 = 2(-f(a) - f'(a)(x-a) + f(x))$$

$$\Leftrightarrow f''(a) = \frac{2(-f(a) - f'(a)(x-a) + f(x))}{(x-a)^2}$$

$x - a = h$

$\Leftrightarrow x = h + a$

$$\Leftrightarrow f''(a) = \frac{-2f(a) - f'(a).h + f(h+a)}{h^2}$$

$$\Leftrightarrow f''(a) = \frac{f(h+a) - 2f(a) - f'(a).h}{h^2}$$

?

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

23. $g(x) = xf(x) \quad g'(x) = f(x) + xf'(x)$

$g''(x) = f'(x) + f'(x) + x.f''(x) = 2f'(x) + xf''(x)$

crescente $g''(0) = 0$

$P_1(x) = g(0) + g'(a)x = f(0)x$

24. $f(x) = \operatorname{arctg}(x^2)$ $\quad f'(x) = \frac{2x}{1-x^4}$ $\quad \forall x \neq \{1, -1\}$

$\mathbb{R} \to]0, \pi/4[$

		-1		0		1	
$1-x^4$	$-$	0	$+$	$+$	$+$	0	$-$
$2x$	$-$	$-$	$-$	0	$+$	$+$	$+$
$2x/1-x^4$	$+$	NE	$-$	0	$+$	NE	$-$

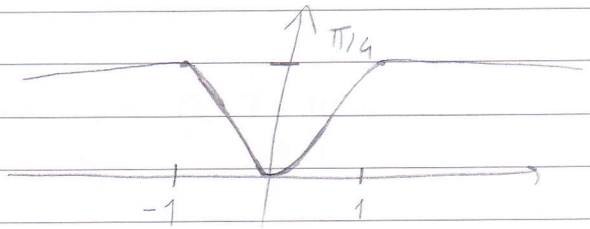
$f(x)$ não tem derivada em $x = \{1, 1\}$

crescente em $]-\infty, -1[$
tem máximo em $x=-1$ $f(x) = \pi/4$
decrescente de $]-1, 0[$
tem um mínimo em $x=0$
crescente em $]0, 1[$
tem máximo em $x=1$ $f(x) = \pi/4$
decrescente em $]1, +\infty[$

$\operatorname{arctg}(-1^2) = \operatorname{arctg} 1 = \pi/4$

$f''(x) = \frac{2(1-x^4) - 2x(4x^3)}{(1-x^4)^2}$

$= \frac{2 - 2x^4 + 8x^4}{(1-x^4)^2} = \frac{6x^4 + 2}{(1-x^4)^2}$



$f''(x) > 0$
$f(x)$ é convexa
não tem pontos de inflexão

25. $f\ \mathbb{R} \to \mathbb{R}$ $\quad f(x) = x^4 e^{-x}$

$f'(x) = 4x^3 \cdot e^{-x} + x^4 \cdot (-e^{-x})$
$= 4x^3 e^{-x} - x^4 e^{-x}$
$= e^{-x}(4x^3 - x^4)$
$= e^{-x} \cdot x^3 (4 - x)$

		0		4	
$e^{-x} \cdot x^3$	$-$	0	$+$	$+$	$+$
$4-x$	$+$	$+$	$+$	0	$-$
0	$-$	0	$+$	0	$-$

$f'(x) = 0$, $x = 0$, $x = 4$

$f(x)$ decrescente em $]-\infty, 0[$
mínimo $x=0$ $\boxed{f(x) = 0}$
crescente em $]0, 4[$
máximo $x=4$ $\boxed{f(x) = 16 \cdot e^{-4}}$
decrescente em $]-\infty, +\infty[$

$+\infty \quad +\infty$

$\lim_{x \to -\infty} x^4 e^{-x} = \lim_{x \to -\infty} +\infty$

$+\infty \cdot 0$

$\lim_{x \to +\infty} x^4 e^{-x} = \lim_{x \to +\infty} \frac{e^{-x}}{\frac{1}{x^4}}$ R. Cauchy

$= \lim_{x \to +\infty} \frac{-e^{-x}}{\frac{3x}{x^8}} = \lim_{x \to +\infty} \frac{-e^{-x} \cdot x^8}{3x}$

26. $f(x) = \frac{\operatorname{sen} x}{1 - \operatorname{sen} x}$

$f'(x) = \frac{\cos x (1 - \operatorname{sen} x) - \operatorname{sen} x (-\cos x)}{(1 - \operatorname{sen} x)^2}$

$= \frac{\cos x - \cos x \operatorname{sen} x + \operatorname{sen} x \cos x}{(1 - \operatorname{sen} x)^2}$

$(\operatorname{sen} x^2)' = 2\operatorname{sen} x \cdot \cos x$

Ficha 10

1.

a) $P\, 2x^2 + 3x^3 = 2P\, x^2 + 3P\, x^3 = 2\frac{x^3}{3} + 3\frac{x^4}{4}$

b) $\frac{x^{-1/2}+1}{-\frac{1}{2}+1} + \log|x| + \frac{x^{-2+1}}{-2+1} = 2\sqrt{x} + \log|x| - \frac{1}{x}$

c) $P\left(\frac{x^2-x+1}{\sqrt{x}}\right) = P\left(x\sqrt{x} - \sqrt{x} + \frac{1}{\sqrt{x}}\right) =$ ($x\sqrt{x} = x^{3/2}$)

$= \frac{x^{3/2+1}}{3/2+1} - \frac{x^{1/2+1}}{1/2+1} + \frac{x^{-1/2+1}}{-1/2+2}$

d) $P\sqrt[3]{1-x} = P(1-x)^{1/3} = \frac{(1-x)^{1/3+1}}{1/3+1} = -\frac{(1-x)^{4/3}}{4/3}$

$\frac{2}{3} - \frac{3}{3} = -\frac{1}{3}$

e) $P\frac{\sqrt[3]{x^2} + \sqrt{x^3}}{x} = P\frac{(x^2)^{1/3} + (x^3)^{1/2}}{x} = P\frac{x^{2/3}}{x} + \frac{x^{3/2}}{x} =$

$\frac{3}{2} - \frac{2}{2} = \frac{1}{2}$

$= P(x^{-1/3} + x^{1/2}) = \frac{3}{2}\cdot x^{2/3} + \frac{2}{3}x^{3/2}$

f) $P\, 2x\sqrt[5]{x^2-1}$ (x^2-1) $= P\,(x^2-1)'\,(x^2-1)^{1/5} = \frac{(x^2-1)^{1/5+1}}{1/5+1}$

$-\frac{1}{3} + \frac{3}{3} = \frac{2}{3}$

$= -\frac{5}{6}\sqrt[5]{(x^2-1)^6} = \frac{5}{6}(x^2-1)\sqrt[5]{(x^2-1)}$

$\frac{1}{2} + \frac{2}{2} = \frac{3}{2}$

g) $\frac{1}{4}P\frac{4x^3}{3+x^4} = \frac{1}{4}\log|3+x^4| = \frac{1}{4}\log(3+x^4)$

h) $\frac{1}{2}P\frac{2e^x}{1+2e^x} = \frac{1}{2}\log|1+2e^x|$

i) $P\frac{\cos x}{1+\operatorname{sen} x} = \log|1+\operatorname{sen} x|$

j) $P\operatorname{sen} 2x = -\frac{\cos 2x}{2}$

k) $\int \frac{\operatorname{sen} 2x}{1+\operatorname{sen}^2 x} = \int \frac{2\operatorname{sen} x \cos x}{1+\operatorname{sen}^2 x} = \log|1+\operatorname{sen}^2 x|$

l) $\int \cos^2 x = \frac{\cos x^3}{-3\operatorname{sen} x}$

$\cos^2 x = \frac{1+\cos 2x}{2}$

m) $\int \frac{1}{\cos^2 x} = \int \frac{1}{\frac{1+\cos 2x}{2}} = \int \frac{2}{1+\cos 2x}$

n) $\int \frac{e^{\operatorname{tg} x}}{\cos^2 x} =$

o) $\frac{1}{2}\int 2x \cos(x^2+2) = \frac{\operatorname{sen}(x^2+2)}{2}$

p) $\int e^x \operatorname{sen}(e^x) = -\cos(e^x)$

q) $\int x^2 \sqrt[3]{1+x^3} = \int x^2 (1+x^3)^{1/3} = \frac{(1+x^3)^{1/3+1}}{1/3+1} =$

$= \frac{(1+x^3)^{4/3}}{4/3} = \frac{3(1+x^3)^{4/3}}{4}$

r) $\int \frac{e^x}{(1+e^x)^2} = \frac{\log(1+e^x)^2}{2}$

s) $\int \frac{\operatorname{sen} x}{1+\cos^2 x} = \operatorname{arctg}(\cos x)$

t) $\frac{1}{2} P \frac{2 \cdot 1}{\sqrt{1-4x^2}} = \arcsin(2x)$

u) $P \frac{x+1}{\sqrt{1-x^2}} = P \frac{x}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}} = P \frac{x}{(1-x^2)^{1/2}} + (\arccos x)'$ ($\arccos x$)

$= \frac{1}{2} P 2x \cdot (1-x^2)^{-1/2} + (\arccos x)' = \frac{(1-x^2)^{1/2}}{\frac{1}{2}} + \arccos x =$

$\frac{-2}{1}$

$= -\sqrt{1-x^2} + \arccos x$

v) $\frac{1}{6} P \frac{6x^3}{(1+x^4)^2} = \frac{\log|(1+x^4)^2|}{6}$

w) $P \cos^3 x \sqrt{\operatorname{sen} x} = P \cos^3 x \cdot (\operatorname{sen} x)^{1/2}$

x) $P \operatorname{tg}^2 x = \frac{\operatorname{tg} x^3}{2}$

k') $P \frac{\cos x}{1+\operatorname{sen}^2 x} = \operatorname{arctg}(\operatorname{sen} x)$

2.

a) $P (x^2+1)^3$ ($2x$) $= \frac{(x^2+1)^4}{4}$

b) $P e^{x+3} = P e^x \cdot e^{x+2} =$

c) $P 2^{x-1} =$

d) $\int \frac{1}{\sqrt[5]{1-2x}} = \int \frac{1}{(1-2x)^{1/5}} = -\frac{1}{2}\int -2(1-2x)^{-1/5} = \frac{(1-2x)^{-\frac{1}{5}+1}}{-1/5+1} \cdot -\frac{1}{2}$

$= \frac{(1-2x)^{4/5}}{4/5} \cdot \left(-\frac{1}{2}\right) = \frac{5(1-2x)^{4/5}}{-8}$

e) $\frac{1}{2}\int \frac{2x}{1+x^2} = \frac{\log|1+x^2|}{2}$

f) $\int \frac{x^3}{x^8+1}$

g) $\int \operatorname{cotg} x = \int \frac{\cos x}{\operatorname{sen} x} = \log|\operatorname{sen} x|$

h) $\int 3^{\operatorname{sen}^2 x} \operatorname{sen} 2x = \int 3^{\operatorname{sen}^2 x} \cdot 2\operatorname{sen} x \cdot \cos x$

$= \int 3^{\operatorname{sen}^2 x} \cdot (\operatorname{sen} x^2)'$

i) $\int \frac{\operatorname{tg}\sqrt{x}}{\sqrt{x}} = \int \operatorname{tg}(x^{1/2}) \cdot x^{-1/2} = \int \frac{\operatorname{sen}(x^{1/2})}{\cos(x^{-1/2})} \cdot x^{-1/2}$

$= \log|-\cos\sqrt{x}|$

j) $\int \frac{e^x}{\sqrt{1-e^{2x}}} = -\operatorname{arccos} e^x$

k) $\int \frac{x}{(1+x^2)^{\alpha}} = \frac{1}{2}\int 2x \cdot (1+x^2)^{-\alpha} = \frac{(1+x^2)^{-\alpha+1}}{2 \cdot (-\alpha+1)}$

l) $P\cos x\cos 2x = P\cos x.(\cos^2 x - \operatorname{sen}^2 x) =$

$= P\cos x.\cos^2 x - \cos x \operatorname{sen}^2 x = P\cos x(1-\operatorname{sen}^2 x) - \cos x \operatorname{sen}^2 x$

$= P\cos x - \cos x \operatorname{sen}^2 x - \cos x \operatorname{sen}^2 x =$

$= P\cos x - P\cos x \operatorname{sen}^2 x - P\cos x \operatorname{sen}^2 x =$

$= \operatorname{sen} x - \frac{2\operatorname{sen}^3 x}{3}$

m) $P\ \operatorname{sen}^3 x \cos^4 x = P\ \operatorname{sen} x(1-\cos^2 x)\cos^4 x =$

$= P(\operatorname{sen} x - \cos^2 x \operatorname{sen} x)\cos^4 x = P\ \operatorname{sen} x \cos^4 x - \cos^6 x \operatorname{sen} x$

$= -\frac{\cos^5}{5} + \frac{\cos^7 x}{7}$

n) $P\ \operatorname{tg}^3 x + \operatorname{tg}^4 x = P\left(\frac{\operatorname{sen} x}{\cos x}\right)^3 + \left(\frac{\operatorname{sen} x}{\cos x}\right)^4$

$= P\ \frac{\operatorname{sen}^3 x}{\cos^3 x} + \frac{\operatorname{sen}^4 x}{\cos^4 x} = P\ \frac{\operatorname{sen} x.(1-\cos^2 x)}{\cos^3 x} +$

$+ P\ \frac{\operatorname{sen}^2 x(1-\cos^2 x)}{\cos^4 x}$

3. a) $P\ \sqrt{2x} + \sqrt{\frac{x}{2}} = 2P(2x)^{1/2} + \frac{1}{2}P\left(\frac{x}{2}\right)^{1/2} =$

$= \frac{4}{3}x^{3/2} + \frac{1}{2}\cdot\frac{2}{3}x^{3/2} = \frac{4}{3}x^{3/2} + \frac{2}{6}x^{3/2}$

$\frac{2}{3}+\frac{1}{2} = \frac{3}{2}$

$x^a =$

3b) $\int 3\sin x + 2x^2 = 3\int \sin x + \frac{2}{3}\int 3x^2$

$= 3.\cos x + \frac{2}{3}.x^3$

c) $\frac{1}{3}\int \frac{3x^2}{1+x^3} = \frac{\log|1+x^3|}{3}$

d) $\frac{1}{-2}\int -2x e^{-x^2} = \frac{e^{-x^2}}{-2}$

e) $\int \frac{3\sin x}{(1+\cos x)^2}$

f) $\int x\sqrt{1+x^2} = \frac{1}{2}\int 2x.(1+x^2)^{1/2} = \frac{(1+x^2)^{1/2+1}}{1/2+1}.\frac{1}{2}$

g) $\frac{1}{2}\int e^{2\sin x} 2\cos x = \frac{e^{2\sin x}}{2}$

h) $\int \frac{1}{1+e^x} =$

i) $\int tgx = \int \frac{\sin x}{\cos x} = -\log|\cos x|$

j) $\int \frac{1}{2+x^2} =$

k) $\int tgx \cdot \sec^3 x = \int \frac{\operatorname{sen} x}{\cos x} \cdot \frac{1}{\cos x^3} =$

$= \int \frac{\operatorname{sen} x}{\cos x} \cdot \frac{1}{\cos x (1 - \operatorname{sen} x^2)} =$

l) $\int \cos^3 x \cdot \operatorname{sen}^3 x = \int \cos^3 x \cdot \operatorname{sen} x (1 - \cos^2 x)$

$= \int \operatorname{sen} x \cos x^3 - \operatorname{sen} x \cos x = -\frac{\cos x^4}{4} + \cos x$

m) $\int \frac{1}{(1+x^2) \operatorname{arctg} x} = \int \left(\frac{1}{1+x^2} \cdot \frac{1}{\operatorname{arctg} x} \right) = \log |\operatorname{arctg} x|$

n) $\int \frac{x}{1+x^4} = \frac{\operatorname{arctg}(x^2)}{2}$

o) $\int \frac{1}{\sqrt{x}(1+x)} = \int \frac{1}{x^{1/2}(1+x)}$

5.

a) $\frac{1}{2} \int 2x \operatorname{sen}(x^2) = -\frac{1}{2} \cos x^2 + C \quad C \in \mathbb{R}$

$f(0) = 0$

$-\frac{1}{2} \cdot \cos 0^2 + C = 0 \Leftrightarrow C = \frac{1}{2}$

b) $\lim_{x \to +\infty} f(x) = 0$ NE

$\cos x$ periódico

---

a) $\int \frac{e^x}{2+e^x} = \log|2+e^x| + C$

$\log|2+e^0| + C = 0 \Leftrightarrow C = \log 3$

b) $\lim_{x \to +\infty} f(x) = 0$ NE

---

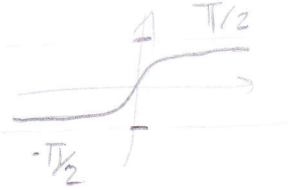
a) $\int \frac{1}{(1+x^2)(1+\operatorname{arctg}^2 x)} = \int \frac{1}{1+x^2} \cdot \frac{1}{1+\operatorname{arctg}^2 x} =$

$= \operatorname{arctg}(\operatorname{arctg} x) + C$

$\operatorname{arctg}(\operatorname{arctg} 0) + C = 0 \Leftrightarrow 0 + C = 0 \Leftrightarrow C = 0$

b) $\lim_{x \to +\infty} f(x) \operatorname{arctg}(\operatorname{arctg} x) = \lim_{y \to \pi/2} \operatorname{arctg} x = \operatorname{arctg} \pi/2 \neq 0$

$x = \operatorname{arctg} \frac{\pi}{2} \Leftrightarrow \operatorname{tg} x = \frac{\pi}{2}$



$$\lim_{x \to +\infty} f(x) = 0 \iff \lim_{x \to +\infty} \operatorname{arctg} \frac{\pi}{2} + C = 0 \iff$$

$$\iff C = -\operatorname{arctg} \frac{\pi}{2}$$